

Structure from Observation:

Probability, Reconstruction, and Finite-Sample Certification

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Abstract

This monograph develops a unified programme showing that probability, dynamics, reconstruction, and their finite-sample signatures are not assumptions but consequences of coherent structured observation.

Part I answers the question: when does observational coherence determine probability? A coherent family of observations on a directed system of observable algebras extends to a unique σ -additive probability measure exactly when it satisfies collective exhaustion (CE) — the condition ruling out persistent mass on globally vanishing events. Two independent routes establish this: Carathéodory extension on the realization space, and compactness of the Stone space. CE is shown to be irreducible: no first-order condition on the query system implies it.

Part II answers the question: given probability, what does the observable measure already contain? The CE-derived probability measure already contains, without further assumption, a conditional regularity structure: the Rokhlin disintegration forces a Markov kernel $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$ that is not chosen but read off. Indexing over time, temporal coherence forces Chapman–Kolmogorov and yields a Markov semigroup and Koopman–Perron duality. Specialising the minimal sufficient factor to a measure-preserving system yields the delay map, and the reconstruction question is internal: whether the factor is faithful. The answer is a three-way equivalence, with the density bridge lemma as the key. When reconstruction holds, the Stone space of Part I is identified with the state space itself.

Part III answers the question: given reconstruction as the asymptotic notion, what can finite data certify? Three theorems — the Algebra Theorem, the Dynamics Theorem, and the Conjunction Theorem — provide computable witnesses for reconstruction, optimal convergence rates, and the result that for deterministic systems all three witnesses certify the same object.

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Part I

When Does Observational Coherence Determine Probability?

Abstract

When does observationally coherent data determine a genuine probability measure? The setting is a directed system of Boolean algebras with surjective refinement maps — a *query system* — carrying a compatible family of finitely additive charges on the cylinder algebra. Coherence (agreement across levels) and finite additivity are purely finitary conditions; the question is what further condition licenses the passage to σ -additivity.

The answer is *collective exhaustion* (CE): a compatible charge family extends to a unique σ -additive measure on the observable σ -algebra if and only if, whenever a decreasing sequence of events vanishes globally, there exists a level at which the mass converges to zero. CE is not derivable from any structural condition on the query system — the obstruction is located by the finite-cofinite counterexample and a model-theoretic argument via Łoś’s theorem — and the same obstruction already appears in the first-order language of Boolean algebras with finitely additive charge, where countable additivity is likewise not first-order axiomatizable. Some extra structure is logically unavoidable; CE is one explicit, minimal way of supplying it.

Two constructions realise the extension under CE. A direct *Carathéodory construction* assumes compatible σ -additive marginals and builds the global measure without topological assumptions. A *Stone construction* starts from finite additivity alone: compactness of the Stone space supplies σ -additivity on the compact completion, and CE appears as the support condition ensuring the measure descends to the realised instantiation space. The constructions differ by where the missing σ -additive structure is supplied — internally by countable reach, or externally by completion — and whenever both apply, they yield the same measure.

The probability measure produced here is the starting point for Papers II and III of this series, which show that it already contains a conditional regularity structure and a dynamical semigroup, and ask what finite data can certify about them.

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1 Introduction

Observable distinctions are naturally encoded by Boolean algebras, and the study of Boolean algebras underpins modern measure theory and its probabilistic extensions. The directed nature of observation — wherein finer levels of resolution refine coarser ones — has, by contrast, played a largely organisational role. In the foundational treatments of Kolmogorov [20] and Choksi [7], the global σ -algebra emerges as the projective limit of compatible measure spaces; directed structure orders the marginals and licenses their assembly, but is not itself interrogated as a source of probabilistic constraint.

This paper interrogates that structure directly. We work with a directed system of Boolean algebras connected by surjective refinement maps, carrying a compatible family of finitely additive charges: charges at different levels agree on events seen from both. Structural coherence and valuation compatibility are purely finitary conditions. The question is what further condition on the valuations is required for finitary data to support a genuine probability measure on the observable σ -algebra.

The condition is *collective exhaustion* (CE): a compatible charge family extends to a σ -additive measure if and only if, whenever a decreasing sequence of events vanishes globally, there exists a level at which the sequence's mass converges to zero. CE rules out purely finitely additive charges, which assign persistent mass to sequences that vanish in the limit; the Yosida–Hewitt decomposition [38] makes this precise by splitting any bounded charge uniquely into a σ -additive part and a purely finitely additive part, with CE equivalent to the vanishing of the latter. It is the exact admissibility condition separating observational coherence from genuine probability: a coherent family of charges can be extended to a σ -additive probability measure if and only if it satisfies CE. A family satisfying CE determines a unique probability measure on the observable σ -algebra. No first-order structural condition can substitute for it — the obstruction is not an artifact of the query-system formalism but already appears in the more primitive language of Boolean algebras with finitely additive charge, where countable additivity is likewise not first-order axiomatizable.

2 Observational structure and compatible charges

2.1 Observational structure

Rather than a single σ -algebra of events, we work with a directed system of Boolean algebras representing observable distinctions at increasing resolution; the global σ -algebra arises as their completion.

Definition 2.1 (Observational structure). An *observational structure* consists of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathcal{O}_i, \mathcal{O}_i)$, the *outcome space at level i* ;
- for each $i \leq j$ in ι , a measurable surjection $\pi_{ij} : \mathcal{O}_j \rightarrow \mathcal{O}_i$, the *refinement map*,

satisfying $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$ for all $i \leq j \leq k$.

Definition 2.2 (Instantiation; query system). An *instantiation* of an observational structure $(\iota, \leq, \{(\mathcal{O}_i, \mathcal{O}_i)\}, \{\pi_{ij}\})$ is a set Ω together with maps $\text{eval}_i : \Omega \rightarrow \mathcal{O}_i$ for each $i \in \iota$, the *evaluation maps*, satisfying the *coherence condition* $\pi_{ij} \circ \text{eval}_j = \text{eval}_i$ for all $i \leq j$. A *query system* is an observational structure together with an instantiation.

2.2 Canonical instantiation and observables

Every observational structure admits a canonical instantiation: the space of all coherent assignments of outcomes across levels. This identifies Ω with the set of all globally consistent observation patterns. Explicitly, the canonical instantiation is the projective limit

$$\Omega = \varprojlim_{i \in \iota} \mathcal{O}_i = \left\{ \omega \in \prod_{i \in \iota} \mathcal{O}_i \mid \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \leq j \right\},$$

with $\text{eval}_i(\omega) = \omega_i$ and $\omega \in \Omega$ generic.

Example 2.3 (Bernoulli observations). Let ι be the set of finite subsets of $\mathbb{N}$, ordered by inclusion. For $i \in \iota$, set $\mathcal{O}_i = \{0, 1\}^{|i|}$ with its discrete σ -algebra $\mathcal{O}_i$, and for $i \subseteq j$ let $\pi_{ij} : \mathcal{O}_j \rightarrow \mathcal{O}_i$ be the coordinate projection. An instantiation assigns to each ω a consistent family of finite binary observations; the canonical instantiation identifies Ω with $\{0, 1\}^{\mathbb{N}}$. Cylinder sets correspond to fixing finitely many coordinates, and a compatible family $\{\nu_i\}$ is precisely the system of finite-dimensional distributions of a Bernoulli process.

Definition 2.4 (Common coarsenings). The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$.

Definition 2.5 (Cylinder sets). For $i \in \iota$ and $A \in \mathcal{O}_i$, the *level- i cylinder with base A* is

$$\text{Cyl}(i, A) = \{\omega \in \Omega : \text{eval}_i(\omega) \in A\}.$$

The *cylinder family* is $\mathcal{C} = \{\text{Cyl}(i, A) : i \in \iota, A \in \mathcal{O}_i\}$, and the *observable σ -algebra* is $\sigma(\mathcal{C})$.

Lemma 2.6 (Cylinder π -system). *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π -system: it is closed under finite intersections.*

Proof. The coherence condition gives $\text{Cyl}(i, A) = \text{Cyl}(j, \pi_{ij}^{-1}(A))$ for $i \leq j$: the same subset of Ω is a cylinder at any finer level. Given $\text{Cyl}(i, A)$ and $\text{Cyl}(j, B)$, let k be a common lower bound. Then

$$\text{Cyl}(i, A) \cap \text{Cyl}(j, B) = \text{Cyl}(k, \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. $\square$

For each $i \in \iota$, let B_i denote the Boolean algebra of level- i cylinders $\{\text{Cyl}(i, A) : A \in \mathcal{O}_i\}$; then $\mathcal{C} = \bigcup_{i \in \iota} B_i$. For $i \leq j$, define the *connecting map*

$$\varphi_{ij} : B_i \rightarrow B_j, \quad \varphi_{ij}(\text{Cyl}(i, A)) = \text{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

2.3 Compatible charges

Definition 2.7 (Compatible family). A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \rightarrow [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i \in \iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \quad \text{for all } i \leq j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on $\mathcal{C}$ by $\mu(\text{Cyl}(i, A)) := \mu_i(\text{Cyl}(i, A))$; compatibility ensures this is well-defined.

Both constructions require the following condition on the query system.

Definition 2.8 (Surjective Evaluation). The query system satisfies *Surjective Evaluation* if $\text{eval}_i : \Omega \rightarrow \mathcal{O}_i$ is surjective for every $i \in \iota$.

Surjective evaluation ensures every outcome at every level is realised by some $\omega \in \Omega$.

Lemma 2.9 (Surjectivity and injectivity from Surjective Evaluation). *If the query system satisfies Surjective Evaluation, then for every $i \leq j$:*

- (i) *the refinement map $\pi_{ij} : \mathcal{O}_j \rightarrow \mathcal{O}_i$ is surjective;*
- (ii) *the connecting map $\varphi_{ij} : B_i \rightarrow B_j$, defined by $\varphi_{ij}(\text{Cyl}(i, A)) = \text{Cyl}(j, \pi_{ij}^{-1}(A))$, is an injective Boolean algebra homomorphism.*

Proof. For (i): given $y \in \mathcal{O}_i$, surjectivity of eval_i yields $\omega \in \Omega$ with $\text{eval}_i(\omega) = y$. Then $\pi_{ij}(\text{eval}_j(\omega)) = \text{eval}_i(\omega) = y$ by the coherence condition, so π_{ij} is surjective.

For (ii): φ_{ij} is a Boolean homomorphism by construction. If $\varphi_{ij}(\text{Cyl}(i, A)) = \varphi_{ij}(\text{Cyl}(i, A'))$, then $\pi_{ij}^{-1}(A) = \pi_{ij}^{-1}(A')$ in $\mathcal{O}_j$. Applying π_{ij} and using surjectivity gives $A = A'$, so φ_{ij} is injective. $\square$

3 Direct extension from compatible probability marginals

Given compatible σ -additive probability marginals, Carathéodory's theorem yields a global measure directly on the instantiation space. The input is a query system satisfying surjective evaluation, sequential common refinements, and common coarsenings, together with a compatible family of σ -additive probability marginals.

3.1 Observational determination

Proposition 3.1 (Observational determination). *Let the query system admit common coarsenings and let P, P' be probability measures on $(\Omega, \sigma(\mathcal{C}))$. If $(\text{eval}_i)_\# P = (\text{eval}_i)_\# P'$ for all $i \in \iota$, then $P = P'$.*

Proof. The equality of pushforward laws implies P and P' agree on every cylinder event. By Lemma 2.6 the cylinder family $\mathcal{C}$ is a π -system generating $\sigma(\mathcal{C})$. The π - λ theorem gives $P = P'$. $\square$

3.2 Cylinder content

Definition 3.2 (Common refinements). The index set $(\iota, \leq)$ admits *common refinements* if every pair $i, j \in \iota$ has an upper bound: there exists $k \geq i$ and $k \geq j$.

The cylinder family may represent the same event at multiple levels: $\text{Cyl}(i, A)$ and $\text{Cyl}(j, B)$ coincide in Ω whenever A and B are preimages of a common set at a finer level. Assigning a numerical value to such an event is well-defined only if the two representations agree — and verifying this requires passing to a common refinement of i and j .

Lemma 3.3 (Finite observational content). *Assume the query system admits common coarsenings and common refinements and satisfies Surjective Evaluation, and let $\{\nu_i\}_{i \in \iota}$ be a family with each $\nu_i \in \mathcal{P}(\mathcal{O}_i)$ compatible under the refinement maps, i.e. $(\pi_{ij})_\# \nu_j = \nu_i$ for all $i \leq j$. Then the assignment*

$$\mu_0(\text{Cyl}(i, A)) := \nu_i(A), \quad A \in \mathcal{O}_i,$$

defines a well-defined finitely additive content on $\mathcal{C}$.

Proof. Well-definedness. Suppose $\text{Cyl}(i, A) = \text{Cyl}(j, B)$. Use common refinements to find k with $k \geq i$ and $k \geq j$. The constraint sets in $\mathcal{O}_k$ are $\pi_{ik}^{-1}(A)$ and $\pi_{jk}^{-1}(B)$. The cylinder equality and surjective evaluation (surjectivity of eval_k) imply these coincide in $\mathcal{O}_k$. Compatibility then gives $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$.

Finite additivity. For disjoint cylinders at possibly different levels, compress to a common refinement k and use additivity of ν_k . $\square$

3.3 Carathéodory extension theorem

Definition 3.4 (Sequential common refinements). The index set $(\iota, \leq)$ admits *sequential common refinements* if every countable family $\{i_n\}_{n \geq 1} \subset \iota$ has an upper bound in ι . Sequential common refinements imply common refinements.

Finite additivity of the cylinder content is not sufficient for Carathéodory's theorem: one needs σ -subadditivity, which requires compressing a countable cylinder cover $\{\text{Cyl}(i_n, A_n)\}$ to a single level. Common refinements handle finite families; sequential common refinements extend this to countable ones, supplying the index-set reach the proof requires.

Theorem 3.5 (Carathéodory observational extension). *Suppose:*

- (i) $(\iota, \leq)$ admits common coarsenings and sequential common refinements;

(ii) the query system satisfies *Surjective Evaluation*;

(iii) each ν_i is a probability measure on $(\mathcal{O}_i, \mathcal{O}_i)$, and the family is compatible under the refinement maps: $(\pi_{ij})_{\#} \nu_j = \nu_i$ for all $i \leq j$.

Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that $(\text{eval}_i)_{\#} P = \nu_i$ for every $i \in \iota$.

Proof. Step 1: Finite content. Lemma 3.3 gives a well-defined finitely additive content μ_0 on $\mathcal{C}$.

Step 2: σ -subadditivity. Let $\text{Cyl}(i, B) \subseteq \bigcup_{n \geq 1} \text{Cyl}(i_n, A_n)$. Apply sequential common refinements to $\{i, i_1, i_2, \dots\}$ to find a single index k with $k \geq i$ and $k \geq i_n$ for all n . Compress each cylinder to level k :

$$\text{Cyl}(i, B) = \text{Cyl}(k, E), \quad \text{Cyl}(i_n, A_n) = \text{Cyl}(k, E_n),$$

where $E = \pi_{ik}^{-1}(B)$ and $E_n = \pi_{i_n k}^{-1}(A_n)$. Surjective evaluation and the cylinder inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in $\mathcal{O}_k$. Therefore

$$\mu_0(\text{Cyl}(i, B)) = \nu_k(E) \leq \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\text{Cyl}(i_n, A_n)),$$

using σ -subadditivity of the single measure ν_k .

Step 3: Carathéodory extension. The family $\mathcal{C}$ is a set semiring generating $\sigma(\mathcal{C})$. It is closed under finite intersections by Lemma 2.6. Moreover, if $E = \text{Cyl}(i, A)$ and $F = \text{Cyl}(j, B)$, choose k with $k \geq i, j$; then

$$E = \text{Cyl}(k, \pi_{ik}^{-1}(A)), \quad F = \text{Cyl}(k, \pi_{jk}^{-1}(B)),$$

so

$$E \setminus F = \text{Cyl}(k, \pi_{ik}^{-1}(A) \setminus \pi_{jk}^{-1}(B)) \in \mathcal{C}.$$

Thus $\mathcal{C}$ is closed under relative differences, confirming the semiring property. Since μ_0 is a finitely additive content on $\mathcal{C}$ that is σ -subadditive for countable covers by cylinder sets (Step 2), the semiring form of Carathéodory's extension theorem yields a measure P on $(\Omega, \sigma(\mathcal{C}))$ extending μ_0 (Bogachev [5], Vol. I, Corollary 1.11.9).

Step 4: Marginal recovery and normalization. For any $A \in \mathcal{O}_i$, $P(\text{Cyl}(i, A)) = \mu_0(\text{Cyl}(i, A)) = \nu_i(A)$, so $(\text{eval}_i)_{\#} P = \nu_i$. Setting $A = \mathcal{O}_i$ gives $P(\Omega) = 1$.

Uniqueness. Two probability measures with the same evaluation marginals agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by Proposition 3.1. $\square$

Remark 3.6 (Relation to Kolmogorov's extension theorem). Kolmogorov's theorem [20] constructs a probability measure on S^T from consistent finite-dimensional distributions. Theorem 3.5 plays the same role for query systems: the instantiation $\Omega = \varprojlim \mathcal{O}_i$ replaces the product S^T , compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure P is built directly on the measurable structure of Ω by Carathéodory, without any appeal to tightness or compactness.

4 Collective Exhaustion

Compatible σ -additive marginals extend directly to a global observable measure. The obstruction to passing from finite additivity to σ -additivity is the possibility that mass persists along a decreasing sequence of events whose intersection is empty.

Example 4.1 (Finite-cofinite obstruction). Let $\mathbb{N}$ carry the finite-cofinite algebra $\mathcal{E}$ where $A \in \mathcal{E}$ if A is finite or A^c is finite. Define a normalized charge by $\mu_0(A) = 0$ if A is finite and $\mu_0(A) = 1$ if A is cofinite.

Disjoint sets in $\mathcal{E}$ are either both finite or one finite and one cofinite (two disjoint cofinite sets cannot exist), so finite additivity holds.

The sequence $E_n = \{n, n+1, \dots\}$ satisfies $E_n \downarrow \emptyset$, yet each E_n is cofinite, so $\mu_0(E_n) = 1$ for all n : the events shrink to nothing but the mass does not drain away. No σ -additive extension exists.

The admissibility condition that rules out this failure is:

Definition 4.2 (Collective Exhaustion). A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_1 \supseteq E_2 \supseteq \dots$ in $\mathcal{C}$ and $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \rightarrow 0$.

CE is a valuation-layer condition: it constrains how charges behave, not the Boolean-algebraic index structure.

We work first at the level of an abstract directed system of Boolean algebras, writing $\mathcal{E}_i$ for the algebra at level i and ℓ_i for its charge; the query-system interpretation returns in §5.

4.1 Single-algebra extension

Definition 4.3 (Boolean charge space). A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where $\mathcal{E}$ is a Boolean algebra of subsets of a set O and $\ell : \mathcal{E} \rightarrow [0, 1]$ is a normalized finitely-additive valuation. No σ -algebra and no topology are assumed on O .

By Stone's representation theorem [36], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ -algebra of $S(\mathcal{E})$. The sets in $\sigma(\mathcal{E}) \setminus \mathcal{E}$ are Baire events that countable operations can reach but no single observable distinction can name.

Lemma 4.4 (Extension criterion). A normalized finitely-additive $\ell : \mathcal{E} \rightarrow [0, 1]$ extends to a σ -additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \dots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \rightarrow 0$. The extension is then unique.

Proof. Standard; see Halmos [14] or Fremlin [11] Vol. 1. Continuity at $\emptyset$ is used in the Carathéodory extension argument to pass from finite additivity to σ -additivity on $\sigma(\mathcal{E})$. $\square$

The condition in Lemma 4.4 is not verifiable within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. In a directed system, one might hope a finer level $j \geq i$ could witness the emptiness level i cannot see.

4.2 Nontrivial refinement

Definition 4.5 (Nontrivial refinement). Let $\pi : O_j \rightarrow O_i$ be a surjective refinement map, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \rightarrow \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$.

Proposition 4.6 (Nontrivial refinement introduces new events). If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. Equivalently, a level i at which $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$ is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.

Proof. Nontriviality asserts the existence of $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$, directly giving the strict inclusion. The contrapositive is the stated equivalence. $\square$

Lemma 4.7 (Strict enlargement under σ -closure). *Let $\mathcal{E}$ be a Boolean algebra of subsets of a countably infinite set O that is not a σ -algebra. Then $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$.*

Proof. If $\mathcal{E}$ is not a σ -algebra, there exists a countable family $\{A_n\} \subset \mathcal{E}$ with $\bigcup_n A_n \notin \mathcal{E}$. By definition $\bigcup_n A_n \in \sigma(\mathcal{E})$, so it lies in $\sigma(\mathcal{E}) \setminus \mathcal{E}$. $\square$

4.3 CE characterises σ -additivity

The directed-system version of CE and the main equivalence are as follows.

Definition 4.8 (Collectively exhaustive family). A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \dots$ in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$.

$\pi_{ij} : O_j \rightarrow O_i$ is the refinement map (j finer than i) and $\pi_{ij}^{-1}(E_n) \subseteq O_j$ is the preimage of E_n at the finer level. No level may assign persistent mass to events the system as a whole sees as empty.

We identify the minimal directed system in which the index-layer hypotheses hold while the finite-cofinite charge still fails σ -additive extension.

Proposition 4.9 (Index-layer conditions do not force σ -additivity). *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ -additive extension at every level.*

Proof. We take the simplest sequentially upper-directed index set: adjoin a single terminal element ω to $\mathbb{N}$, setting $n \leq \omega$ for all $n \in \mathbb{N}$. Take all outcome spaces to be $\mathbb{N}$, all event algebras to be the finite-cofinite algebra $\mathcal{E}$, all refinement maps to be the identity, and each ℓ_i to be the finite-cofinite charge ℓ of Example 4.1. Compatibility holds trivially. The system satisfies every structural condition. By Example 4.1, ℓ is not σ -additive and admits no σ -additive extension; hence every level fails σ -additive extension. $\square$

Thus the obstruction reappears: finite additivity at j does not force continuity at $\emptyset$.

Theorem 4.10 (CE Characterisation). *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

- (i) *The family is collectively exhaustive.*
- (ii) *ℓ_i extends uniquely to a σ -additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.*

Proof. (i) $\Rightarrow$ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$. So ℓ_i is continuous at $\emptyset$. Lemma 4.4 gives the unique σ -additive extension.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every i . Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \rightarrow \mu_i(\emptyset) = 0$ by σ -additivity of μ_i and normalization. Take $j = i$. $\square$

The question “do the structural conditions on a directed system force σ -additivity?” is ill-posed: it conflates conditions on the index structure with conditions on the charges. Proposition 4.9 separates them; Theorem 4.10 identifies the exact valuation-layer condition. Sequential upper-directedness enters only when assembling per-level extensions into a global measure (§3).

4.4 CE is non-derivable

The Yosida–Hewitt decomposition [38] identifies the obstruction Proposition 4.9 has exposed. Every finitely additive probability decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ -additive and ℓ_p is *purely finitely additive*. The finite-cofinite content has $\ell_p = \ell$. By Theorem 4.10, CE holds precisely when $\ell_p = 0$. The obstruction is therefore not a feature of any particular structural hypothesis: it is the persistence of a purely finitely additive component that no first-order structural condition can eliminate.

Proposition 4.11 (CE non-derivability). *Collective exhaustion is not implied by any structural condition on a query system.*

- (i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*
- (ii) (Universal case.) *No condition Φ expressible in the first-order language of query systems implies CE.*

Proof. The particular case is Proposition 4.9: the finite-cofinite charge satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

Step 1. A first-order condition on query systems is preserved under ultraproducts by Łoś’s theorem [24]: if Φ holds in every member of a family of query systems, it holds in their ultraproduct.

Step 2. Since the finite-cofinite construction works on any countably infinite set, we may realise it on $\mathbb{Q}$. Fix an enumeration $q : \mathbb{N} \rightarrow \mathbb{Q}$, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter, and let δ_n be the point mass at $q(n) \in \mathbb{Q}$. Each δ_n is a normalized σ -additive charge satisfying every structural condition including CE. The ultraproduct $\prod_{\mathcal{U}} \delta_n$ assigns to $E \subseteq \mathbb{Q}$ the value $\lim_{\mathcal{U}} \mathbf{1}_{q(n) \in E}$.

If E is finite then $q(n) \in E$ holds for only finitely many n , so the ultralimit is 0. If E is cofinite then $q(n) \in E$ holds for cofinitely many n , so the ultralimit is 1, since $\mathcal{U}$ extends the cofinite filter. The ultraproduct is therefore exactly the finite-cofinite charge ℓ of Example 4.1 on $\mathbb{Q}$.

Step 3. By Step 1, any universally valid first-order condition Φ holds in the ultraproduct whenever it holds in each δ_n . But by Step 2 the ultraproduct is exactly ℓ , and ℓ fails CE. Therefore no first-order structural condition can imply CE. $\square$

The ultrafilter-based charge is the canonical witness: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty.

4.5 Finite coherence does not determine probability

The obstruction in Proposition 4.11 is not an artifact of the query-system language. In the more primitive setting of Boolean algebras with a normalized finitely additive charge, the same conclusion holds: no first-order theory in that language characterizes those algebras whose charge extends to a σ -additive measure. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [26]. The probability logic community has treated this as background knowledge — Hoover [17] requires an infinitary rule to axiomatize σ -additive probability logic, and Fagin, Halpern, and Megiddo [10] state without proof that countable additivity cannot be expressed by any formula in the natural language — but an explicit proof does not appear in the literature. The companion note supplies one via the same ultraproduct mechanism: Dirac charges δ_n satisfy any putative characterizing theory, yet their ultraproduct is the finite-cofinite charge ℓ , which fails σ -additivity, contradicting Łoś’s theorem.

Finite coherence therefore underdetermines probability in a logically precise sense. Foundational disputes about probability are not terminological; they concern which extra structure licenses the passage from finite coherence to probability, and some such structure is unavoidable. CE is the condition adopted here; the companion note [26] gives the full general proof.

Example 4.12 (Propositional probability as a query system). The framework is not specific to empirical measurement. Let T be a consistent countable propositional theory with sentences $\phi_1, \phi_2, \dots$. At level n , let

$$\mathcal{E}_n = \text{Lind}(\phi_1, \dots, \phi_n)$$

be the finite Boolean algebra of T -equivalence classes of Boolean combinations of $\phi_1, \dots, \phi_n$. Its atoms are the maximal consistent truth-value assignments to $\phi_1, \dots, \phi_n$ compatible with T . For $n \leq m$, restricting an assignment on $\phi_1, \dots, \phi_m$ to its first n sentences gives the refinement map π_{nm} . The canonical instantiation is

$$\Omega = \varprojlim_n \text{atoms}(\mathcal{E}_n) = \{\text{complete consistent extensions of } T\},$$

and $\text{eval}_n(\omega)$ is the restriction of a complete extension ω to $\{\phi_1, \dots, \phi_n\}$. Surjectivity of each eval_n follows from the Lindenbaum extension lemma: every maximal consistent assignment to $\phi_1, \dots, \phi_n$ extends to a complete consistent extension of T .

A compatible family of normalised charges $\{\nu_n\}$ on $\{\mathcal{E}_n\}$ is precisely a coherent system of finite-level probability assignments. A decreasing sequence of cylinder events $\text{Cyl}(n_k, A_k)$ with $n_k \nearrow \infty$ has empty intersection in Ω if and only if the corresponding propositional conditions $\alpha_k \in \mathcal{E}_{n_k}$ are jointly unsatisfiable over T : no complete extension satisfies all of them simultaneously. CE requires that the charges drain away in this case,

$$\nu_{n_k}(\alpha_k) \longrightarrow 0,$$

ruling out persistent mass on conditions that collectively vanish. Theorem 4.10 therefore yields a unique σ -additive probability measure on the space of complete consistent extensions of T exactly when the charge family satisfies CE.

Remark 4.13. Under Stone duality, the Stone space of $\varinjlim_n \mathcal{E}_n$ is the space of possible worlds: complete consistent extensions appear as the principal ultrafilters, while non-principal ultrafilters are ideal limit points — internally coherent but not realised by any complete extension of T . By Proposition 5.7, CE is exactly the condition that no mass escapes to these ideal points. CE is thus not a condition tailored to empirical observation; it is the general admissibility principle separating coherent finite-level probability assignments from honest probability on the completed space of possible worlds.

5 Stone realization from finitely additive data

Fortunately σ -additivity can be derived. Starting only from finitely additive charges, compactness of the Stone space produces a σ -additive measure on the compact completion; CE then reappears as the support condition that returns this measure to the realised instantiation. No sequential common refinements are assumed: the σ -additivity that Carathéodory obtains by compressing countable covers within the index set is reproduced here by compactness of the Stone space, without any countable reach into ι .

5.1 Stone duality for the direct limit

Injectivity of the connecting maps (Lemma 2.9(ii)) is the key structural input for the Stone construction; we restate it here for reference.

Lemma 5.1 (Injectivity of connecting maps). *If the query system satisfies Surjective Evaluation, then $\varphi_{ij} : B_i \rightarrow B_j$ is an injective Boolean algebra homomorphism for every $i \leq j$.*

Proof. Lemma 2.9(ii). □

Proposition 5.2 (Stone duality for the direct limit). *The Stone space of the direct limit satisfies $\text{St}(\bigcup_i B_i) \cong \varprojlim_i \text{St}(B_i)$.*

Proof. By Lemma 5.1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [19, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\text{St}(\bigcup_i B_i) \cong \varprojlim_i \text{St}(B_i)$. □

5.2 The inverse limit measure

Proposition 5.3 (Inverse limit measure). *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \text{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \text{St}(B_i) \rightarrow \text{St}(B_j)$ agrees with the Borel measure on $\text{St}(B_j)$ corresponding to μ_j .*

Proof. Step 1: Single-algebra correspondence. For each i , Stone duality identifies B_i with the clopen algebra $\text{Clop}(\text{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\text{St}(B_i)$. Since $\text{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [6], §6, or [14], §§53–54. Call this measure $\hat{\mu}_i$ on $\text{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\text{St}(B_j) \rightarrow \text{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\text{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\text{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \text{St}(B_i)$ is compact Hausdorff. By [7, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \text{St}(B_i))$. □

5.3 Structural identification

The Stone space $\text{St}(\mathcal{C})$ contains Ω as a subspace via the Stone embedding, but only if distinct states are separated by observable events.

Definition 5.4 (Discriminability). The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Discriminability ensures the Stone embedding is injective: without it, distinct states would be identified in the completion, and the pullback of the measure on $\text{St}(\mathcal{C})$ would not recover a well-defined measure on Ω .

Let $\text{pure} : \Omega \rightarrow \text{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \text{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\text{St}(\mathcal{C})$.

Proposition 5.5 (Structural identification). *Suppose the query system discriminates points. Then*

$$\text{pure}^{-1}(\mathcal{B}_{\text{Ba}}(\text{St}(\mathcal{C}))) = \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\text{Ba}}$ denotes the Baire σ -algebra.

Proof. ($\supseteq$): For any $E = \text{Cyl}(i, A) \in \mathcal{C}$, $\text{pure}^{-1}([E]) = \{\omega : E \in \text{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\text{pure}^{-1}(\mathcal{B}_{\text{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \text{pure}^{-1}(\mathcal{B}_{\text{Ba}})$.

($\subseteq$): The Stone space $\text{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [19, I.3.2], so the Baire σ -algebra of $\text{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V , $\text{pure}^{-1}(V) \in \sigma(\{\text{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability (Definition 5.4) implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \text{pure}(\omega)$ but $E \notin \text{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. $\square$

Remark 5.6. On a compact Hausdorff space the Baire σ -algebra is contained in the Borel σ -algebra, with equality when the space is second-countable. We work throughout with the Baire σ -algebra on $\text{St}(\mathcal{C})$; this suffices because the measure $\hat{\mu}$ constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

5.4 CE as support condition

Proposition 5.7 (Support condition). *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$ is supported on $\text{pure}(\Omega)$, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [38, 30], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. Under Stone duality, μ_c corresponds to mass on the principal ultrafilters $\text{pure}(\Omega) \subset \text{St}(\mathcal{C})$, while μ_p corresponds to mass on the non-principal ultrafilters [30], Ch. 10.

CE is equivalent to $\mu_p = 0$: CE requires $\mu(E_n) \rightarrow 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes (Theorem 4.10). Therefore $\hat{\mu}_p = 0$ and $\hat{\mu}$ is supported on $\text{pure}(\Omega)$. $\square$

5.5 The Stone observational extension

Theorem 5.8 (Stone observational extension). *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\text{Cyl}(i, A)) = \mu_i(\text{Cyl}(i, A)) \quad \text{for all } i \in \iota \text{ and } A \in \mathcal{O}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\S 5.2, \text{Step 1}} \{\hat{\mu}_i\} \xrightarrow{\S 5.2, \text{Step 3}} \hat{\mu} \text{ on } \varprojlim_i \text{St}(B_i) \xrightarrow{\text{Prop. 5.5}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.2, $\text{St}(\mathcal{C}) \cong \varprojlim_i \text{St}(B_i)$. By Proposition 5.3, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \text{St}(B_i) \cong \text{St}(\mathcal{C})$.

By Proposition 5.7, $\hat{\mu}$ is supported on $\text{pure}(\Omega)$. Since pure is injective (discriminability), the pushforward $P = (\text{pure})_* \hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω .

By Proposition 5.5, pure^{-1} maps Baire sets in $\text{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \text{Cyl}(i, A)$,

$$P(E) = \hat{\mu}([E]) = \mu_i(\text{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Proposition 3.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to surjective evaluation, discriminability, and CE. $\square$

It may appear that compactness of the Stone space eliminates the need for CE. Compactness yields a σ -additive Baire measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$, not yet on Ω . Descent to Ω is equivalent to $\hat{\mu}$ being supported on $\text{pure}(\Omega)$, the principal ultrafilters; and by Proposition 5.7 this support condition is exactly CE. In the Stone picture, CE is not an additional hypothesis but a *support condition*: it is what ensures the measure concentrates on realised states rather than ideal limit points.

6 Synthesis

The results of this paper separate three layers: coherence, completion, and admissibility. A finitely additive charge gives a coherent valuation on the Boolean algebra of observable distinctions; Stone duality completes that algebra into the space of all coherent distinction patterns, including nonprincipal ones; and collective exhaustion is the condition that determines when such a valuation extends to genuine probability. In this sense, the gap between finite coherence and probability is the gap between unrestricted support on the Stone completion and support that remains honest under observational exhaustion.

The Yosida–Hewitt decomposition makes this precise: $\mu_p = 0$ is both the vanishing of the purely finitely additive part and the condition that no mass escapes to non-principal ultrafilters.

When the index set admits sequential common refinements and the query system discriminates points, both constructions are available simultaneously. The following proposition confirms they produce the same observable measure.

Proposition 6.1 (Uniqueness of the observable measure). *Suppose the query system satisfies Surjective Evaluation, Discriminability, and sequential common refinements and common coarsenings, and let $\{\nu_i\}$ be a compatible family of probability measures. Let P^C be the measure produced by Theorem 3.5 and P^S the measure produced by Theorem 5.8. Then $P^C = P^S$.*

Proof. Both measures agree on every cylinder set: $P^C(\text{Cyl}(i, A)) = \nu_i(A) = P^S(\text{Cyl}(i, A))$. By observational determination (Proposition 3.1), they are equal on $\sigma(\mathcal{C})$. $\square$

Corollary 6.2. *The observable measure may be obtained by two distinct constructions: by countable reach in ι via sequential common refinements (Theorem 3.5), or by the Stone completion of $\mathcal{C}$ (Theorem 5.8). Whenever both constructions apply, they yield the same measure.*

The Stone space makes the geometry of that demand visible. Every point of $\text{St}(\mathcal{C})$ is a maximal consistent assignment of yes/no answers to every observable distinction — a coherent distinction pattern, whether or not any $\omega \in \Omega$ realises it. The realised instantiation sits inside $\text{St}(\mathcal{C})$ as the principal ultrafilters; beyond it lie ideal limit points, internally coherent but unrealised.

The measure now exists. The next question is what it implies: whether prediction, dynamics, and ultimately the recovery of the underlying state space follow from observation alone — or whether they too must be assumed.

Part II

Dynamics and Reconstruction in the Observable Measure

Abstract

Starting from a σ -additive probability measure on the algebra of discriminable events [28], the Rokhlin disintegration theorem forces a conditional regularity kernel and a minimal sufficient factor Q_* , the coarsest reduction carrying full conditional information. Indexing over time, temporal coherence forces the Chapman–Kolmogorov equation, yielding a Markov semigroup $\{\Pi_t\}$ and Koopman–Perron duality; the dynamics of observation and the underlying state space are derived, not assumed. In a measure-preserving system Q_* becomes the delay map Φ_h , and reconstruction — that $\mathcal{O}_h = \mathcal{B} \bmod \mu$, density of the delay algebra in L^2 , and injectivity of Φ_h are equivalent — is the faithfulness question about this factor. The Stone space [28] built to derive σ -additivity is identified with X itself, up to measure-theoretic isomorphism, when reconstruction holds.

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1 Introduction

Koopman studied dynamical systems through observables [21], but the dynamical law T remains primitive: the operator $U_T f = f \circ T$ is defined only after an underlying evolution has already been specified. The classical stochastic-process tradition proceeds similarly. Prediction is formulated relative to a filtration, so temporal order and information flow are built into the formalism from the outset [8]. That ordering is natural, but it need not be conceptually prior. One may instead ask whether observable probability already carries a conditional regularity between observables, and whether the familiar semigroup and predictive structures arise only when that regularity is organised coherently over time.

In [28], σ -additivity is not assumed but derived: a probability measure is forced into existence by the coherence of finite observations, at the cost of a single admissibility condition that no first-order structural requirement can supply. The measure is earned, not given. It does, however, already carry more than it was built to carry. Any two observations Q and F — no hypothesis on time or order — determine a conditional regularity of F given Q : the Markov kernel $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$, which the Rokhlin disintegration theorem [32] forces into existence — it need not be separately posited. Composing with Q gives the minimal sufficient factor Q_* , the coarsest reduction of the sample point that carries full conditional information about F . Temporal prediction is the special case in which F is a future observable; nothing in the construction requires it.

Index the conditional regularity over time and the kernels $\{\Pi_t\}$ must compose consistently. That coherence forces Chapman–Kolmogorov — derived, not assumed — and from it a Markov semigroup and Koopman–Perron duality $\int K_t g d\mu = \int g d(\mu \star \Pi_t)$. In a measure-preserving system with $Q_t = h \circ T^t$, the factor Q_* becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the question becomes internal: does this factor see all of X , or only a proper quotient? The answer — that $\mathcal{O}_h = \mathcal{B} \bmod \mu$, density of the delay algebra in L^2 , and injectivity of Φ_h are all equivalent — is the reconstruction theorem. The Stone space [28] built to derive σ -additivity is identified with X itself when reconstruction holds.

2 Conditional regularity and the canonical factor

2.1 The conditional regularity kernel

Let $(\Omega, \mathcal{F}, P)$ be a σ -additive probability space on an observable algebra [28], and let Q, F be two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [32] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\kappa_Q : \alpha \rightarrow \mathcal{P}(\beta)$ satisfying

$$\int_A \kappa_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Conditional regularity kernel). The *conditional regularity kernel* of F given Q is κ_Q , the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\kappa_Q(q, A) = P(F \in A \mid Q = q)$. When F is a future observable the kernel describes conditional prediction; the object itself is neutral on time.

The kernel is covariant under measurable maps: if $\pi : \alpha \rightarrow \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\kappa_Q(q, \cdot) = \kappa_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$ -almost every q , by uniqueness of disintegration.

Composing κ_Q with Q gives the map sending each sample point to its conditional regularity.

2.2 The minimal sufficient factor

Definition 2.2 (Minimal sufficient factor). The *minimal sufficient factor* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \kappa_Q(Q(\omega), \cdot).$$

Since κ_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Factorization via the canonical factor). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g d\kappa_Q(q, \cdot)$ is strongly measurable; since $Q_* = \kappa_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$ -measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_A \int g d\kappa_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. □

Since $\sigma(Q_*) \subseteq \sigma(Q)$ and the right-hand side is already $\sigma(Q_*)$ -measurable, Q_* is a sufficient statistic for F given Q : $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e. It is *minimal* in the sense that every sufficient statistic factors through it: Q_* is the coarsest reduction of ω that carries the full conditional regularity of F given Q . The operator $(K_{Q_*}g)(q_*) = \int g dq_*$ on $\mathcal{P}(\beta)$ recasts this as $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e.

3 Dynamics and duality

3.1 Markov semigroup

When the conditional regularity kernel κ_Q is indexed over a time parameter, the kernels cannot be chosen independently: the regularity of F at time $t+s$ given Q at time 0 must be consistent with first observing at time s and then at time t . This consistency requirement forces the Chapman–Kolmogorov equation.

Concretely, index over a discrete time horizon $\mathbb{N}$ and let $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ be a Markov kernel on a state space γ , with $\Pi_0(q, \cdot) = \delta_q$. Temporal coherence of the conditional regularity demands

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

— the Chapman–Kolmogorov equation, derived from the structure rather than assumed. A family $\{\Pi_t\}$ satisfying it is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \text{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 3.1 (Semigroup property). *$K_{t+s}g = K_t(K_s g)$ for all bounded measurable g .*

Proof. $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) = \int (\int g d\Pi_t(q', \cdot)) d\Pi_s(q, q') = \int (K_t g)(q') d\Pi_s(q, q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. $\square$

The pushforward of a measure μ on γ through the kernel is $P_t^* \mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$.

3.2 Koopman–Perron duality

Theorem 3.2 (Koopman–Perron duality). *For every bounded measurable g and finite measure μ :*

$$\int (K_t g) d\mu = \int g d(P_t^* \mu).$$

Proof. $\int g d(P_t^* \mu) = \int g d(\mu \star \Pi_t) = \int_{\gamma} \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_t g) d\mu$, by Bochner–Fubini. $\square$

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$, the stochastic picture collapses.

3.3 Specialisation to measure-preserving systems

Proposition 3.3 (Deterministic limit). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*

Proof. $(K_t g)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. $\square$

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g d\mu = \int g d(P_t^* \mu)$ becomes the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t \mu)$.

Example 3.4 (Finite Markov chain). Let $X = \{1, \dots, k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution μ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The conditional regularity kernel of X_1 given X_0 is $\kappa(i, \cdot) = P(i, \cdot)$, the i -th row of P : this is the unique Markov kernel satisfying $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$ for all bounded f . The t -step kernel $\Pi_t(i, \cdot) = P^t(i, \cdot)$ is the corresponding semigroup kernel, and temporal coherence gives $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically σ -additive, so the admissibility condition of [28] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever h separates points of X .

The full structure produced by the construction is the triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$: the canonical factor space, stationary distribution $\nu_{Q_*} = P \circ Q_*^{-1}$, and Markov semigroup. This is the *observable dynamical system* derived from (Q, F, P) : the conditional regularity of F relative to Q , promoted to a dynamical object by temporal coherence.

4 From regularity to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n \omega))_{n \in \mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The canonical factor space $\Phi_h(X)$ is a subset of $\mathbb{R}^\mathbb{Z}$ encoding the dynamical information that h can see. Whether it is a faithful image of X depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X . Injectivity of Φ_h modulo μ -null sets is equivalent to the observable σ -algebra

$$\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$$

equalling $\mathcal{B}(X)$ modulo μ .

Definition 4.1 (Faithful factor). A measurable map $\psi : (X, \mathcal{B}, \mu) \rightarrow (Y, \mathcal{Y})$ is *faithful modulo μ* if

$$\sigma(\psi) = \mathcal{B} \quad \text{modulo } \mu,$$

equivalently, if ψ separates μ -a.e. pair of points of X . The reconstruction question is whether Φ_h is faithful modulo μ .

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable σ -algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$.

4.1 Separation defect and the role of CE

For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \dots \vee \mathcal{E}_{n_k}$ from a directed system $(\mathcal{E}_n)_n$, the *separation defect* is

$$\delta(\mathcal{G}) := \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E),$$

where $\sigma(\mathcal{C})$ is the observable σ -algebra of the full system. It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C}) \bmod \mu$. A sequence $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Theorem 4.2 (CE drives separation defect to zero). *Let $(\mathcal{E}_n)_n$ be a query system with realization measure μ satisfying collective exhaustion [28]. Under any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Proof. Fix $A \in \sigma(\mathcal{C})$. Since $(\mathcal{G}_k)$ is honest, $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$, so A is $\sigma(\bigcup_k \mathcal{G}_k)$ -measurable. By the $L^1(\mu)$ martingale convergence theorem, $\mathbf{1}_{E_k} \rightarrow \mathbf{1}_A$ in $L^1(\mu)$ where $E_k = \mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$ rounded to the nearest atom; in particular $\mu(A \triangle E_k) \rightarrow 0$. CE [28] ensures no mass persists on the vanishing sets $A \triangle E_k$, confirming $\inf_{E \in \mathcal{G}_k} \mu(A \triangle E) \rightarrow 0$ uniformly over A . Hence $\delta(\mathcal{G}_k) \rightarrow 0$. $\square$

Sections 5–6 carry out the reconstruction in the delay-map setting.

Remark 4.3 (The temporal specialisation). With $Q_t = h \circ T^t$, the algebra $\mathcal{O}_h^{(L)}$ is an honest refinement scheme once reconstruction holds (§§5–6). The separation defect specialises to $\delta(L) := \delta(\mathcal{O}_h^{(L)})$, the central quantity of [25]: exponential decay of $\delta(L)$ under mixing yields minimax-optimal certification rates. In the general query setting, exhaustion and mixing are separate conditions; the delay-map setting collapses the first by construction.

5 Density bridge and reconstruction

5.1 Density bridge lemma

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \rightarrow X$ is a measurable, μ -preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 5.1 (Observable algebra). The *observable algebra* generated by h is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 5.2 (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n , the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞ .

Lemma 5.3 (Density bridge). *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ .

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$ -measurable, hence $\mathcal{M}$ -measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V . By the functional monotone class theorem [9], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \rightarrow f$ pointwise a.e. with $f \in L^\infty$, then $f_n \rightarrow f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$ -indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$ -simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B}$ mod μ . $\square$

Remark 5.4 (Role of the algebra structure). The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed linear span $\mathcal{H}_{\mathcal{A}} = \overline{\text{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu = \text{Lebesgue}$, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ -algebra (the dyadic structure separates points). But $\text{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2 . So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B}$ mod μ — see Corollary 6.2.

5.2 Reconstruction theorem

Theorem 5.5 (Reconstruction theorem). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

- (i) $\mathcal{O}_h = \mathcal{B}$ modulo μ ;
- (ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;
- (iii) $\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ modulo μ .

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B}$ mod μ .

(i) $\Leftrightarrow$ (iii). By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. So $\mathcal{O}_h = \mathcal{B}$ mod μ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ mod μ , which is exactly condition (iii). $\square$

Condition (iii) means Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h)_*\mu)$. Since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$, the measure $(\Phi_h)_*\mu$ is shift-invariant, and (X, T, μ) is isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to $\Phi_h(X)$.

6 Cyclic vectors and Stone space identification

6.1 Cyclic vectors

Definition 6.1 (Cyclic vector). The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

Corollary 6.2 (Cyclic implies reconstruction). *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T , then $\mathcal{O}_h = \mathcal{B}$ modulo μ .*

Proof. Since $\mathcal{A}_h$ contains $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. $\square$

Remark 6.3 (The implication is strict). The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2 -spectrum*: $L^2(X, \mu)$ is a cyclic U_T -module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [13] but is not universal.

6.2 Stone space identification

The Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Theorem 6.4 (Stone space identification). *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ . Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. *Injectivity mod μ .* Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\supset x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ -a.e.

σ -algebra identification. The pullback ι^{-1} maps the Baire σ -algebra of $\text{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

Measure recovery. Since μ is σ -additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [28] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space [28] — built as a compact ambient space for deriving σ -additivity — is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

Example 6.5 (Cantor set: reconstruction without smoothness). Let $\mathcal{C} \subset [0, 1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \dots) = \sum_{k \geq 1} d_k/3^k$. Let $S : \mathcal{C} \rightarrow \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0, 2\}^{\mathbb{N}}, \nu, \sigma)$ via π ; in particular S is $\mu_{\mathcal{C}}$ -invariant and ergodic.

For $x = \pi(d_1, d_2, \dots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \dots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in [0, \frac{1}{3}], \quad d_{n+1} = 2 \iff S^n x \in [\frac{2}{3}, 1],$$

so $h \circ S^n$ recovers the $(n+1)$ -st ternary digit of x . The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \dots, S^L x)$ reads off digits $d_1, \dots, d_{L+1}$; its fibres are the generation- $(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k, 1 \leq k \leq L+1\}, \quad z \in \{0, 2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0, 2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$ -a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \rightarrow \infty$:

$$\mathcal{O}_h = \sigma\left(\bigcup_{L \geq 0} \mathcal{O}_h^{(L)}\right) = \mathcal{B}(\mathcal{C}) \mod \mu_{\mathcal{C}}.$$

By Theorem 5.5, Φ_h is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \rightarrow \infty$.

The observable algebra also encodes the geometry of $\mathcal{C}$ directly. At each lag L there are 2^{L+1} fibres of Euclidean diameter $(1/3)^{L+1}$, so the box-counting ratio $\log 2^{L+1} / \log 3^{L+1} = \log 2 / \log 3$ is an algebraic invariant of the algebra itself, constant across all lags. The Hausdorff dimension is not a separate geometric calculation: it is read off from the refinement rate of the observable algebra.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2 / \log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — C^2 ambient dynamics, integer manifold dimension d , embedding bound $2d+1$ — are unavailable before one even asks about regularity of S ; nor do the finite-lag extensions of Sauer–Yorke–Casdagli [33], which still require a finite-dimensional ambient manifold or attractor, apply. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness and finite dimensionality play no role.

The classical Takens embedding theorem [37] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d+1$. Sauer–Yorke–Casdagli [33] extend this to prevalence-generic observables and Lipschitz maps, while retaining the finite-dimensional manifold hypothesis and the finite-lag, finite-dimension embedding framework. The reconstruction theorem here differs in kind: it requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h , the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2 -dense: the density condition holds and the results are compatible. For non-smooth or zero-dimensional systems — the Cantor example being representative — the closer historical ancestor is the symbolic dynamics and generating-partition tradition [1]: representing orbits through successively finer finite partitions is exactly what the observable algebra $\mathcal{O}_h^{(L)}$ does as L grows, with faithfulness of Φ_h replacing the generating-partition condition.

Remark 6.6 (Observable separation and Lyapunov exponents). When T is C^r and reconstruction holds, the delay map Φ_h carries a natural instability rate: for μ -a.e. x and nearby y ,

$$\lambda_h(x, y) := \limsup_{n \rightarrow \infty} \frac{1}{n} \log d(\Phi_h(T^n x), \Phi_h(T^n y)).$$

This *observable separation exponent* measures how fast the reconstructed dynamics distinguishes initial conditions through the observable, rather than how fast tangent vectors grow. When h is generic in the sense of Takens [37], the finite-lag embedding $\Phi_h^{(L)}$ is bi-Lipschitz on the attractor, sandwiching λ_h between the top Lyapunov exponent λ_1 from above and below; one expects $\lambda_h = \lambda_1$ μ -a.e., but making this precise requires comparing the induced metric on $\Phi_h^{(L)}(X)$ with the ambient Riemannian geometry as $L \rightarrow \infty$, which is beyond the scope of the present paper. The quantitative geometry of fibre separation in this setting is taken up in [25].

7 Discussion

The measure derived in [28] already contains, without further assumption, the structures developed here. The Rokhlin disintegration reads off a conditional regularity kernel for any pair of observables; temporal indexing of that kernel forces Chapman–Kolmogorov and hence the Markov semigroup; and in a measure-preserving system the minimal sufficient factor becomes the delay map. Nothing is added to the measure — only consequences are drawn.

The individual components are classical. Rokhlin disintegration [32] gives regular conditional distributions on standard Borel spaces; Halmos and Savage [15] laid the measure-theoretic foundation for sufficient statistics via the Radon–Nikodym theorem; Blackwell [4] placed sufficiency in comparative terms; and Doob [8] established conditional expectation, filtrations, and Markov semigroups as the language of modern probability. Predictive-state representations [23] and observable operator models [18] likewise distrust hidden-state parametrisations and encode state through observable predictions. Here the kernel, the factor, the semigroup, and the delay map are not assumed, chosen, or built from a prior dynamical law — they are consequences of a single measure on an observable algebra, disclosed in sequence.

The reconstruction theorem closes a circle opened in [28]. The Stone space $\text{St}(\mathcal{C})$ built there to establish σ -additivity is an abstract completion of the observable algebra, containing the realised state space as the principal ultrafilters but not yet identified with it. When Φ_h is faithful modulo μ (Definition 4.1), the three equivalent conditions of Theorem 5.5 jointly force $\sigma(\Phi_h) = \mathcal{B}$ modulo μ , and the Stone space collapses onto X itself: the abstract completion is the concrete state space. The question left open in [28] — whether prediction, dynamics, and state recovery follow from observation — is answered affirmatively, under faithfulness.

What faithfulness means in practice, and whether it can be certified from finite data, is taken up in [25].

Part III

Certifying Reconstruction from Finite Data

Abstract

Papers [28, 27] establish that a σ -additive probability measure is forced by observational coherence (collective exhaustion), and that this measure already contains, without further assumption, both the conditional regularity structure and the dynamical semigroup — with reconstruction as the question of whether the minimal sufficient factor is faithful. This paper asks what finite data can certify about those structures.

At the level of a general query system, the *population separation defect* $\delta(\mathcal{G})$ measures how far finite observations fall from full resolution; it vanishes under collective exhaustion and honest refinement. The empirical collision probability is a U-statistic proxy, and a persistent empirical floor is evidence against collective exhaustion — a CE-shadow theorem requiring no temporal or geometric structure.

In the dynamical specialisation, the certification framework meets the reconstruction theorem of [27]: honest refinement, a design condition in the general setting, is here automatic — a consequence of reconstruction — and the remaining degrees of freedom (mixing rate, embedding dimension) are what rate-optimal guarantees exploit. Three theorems certify reconstruction from a time series $x_1, \dots, x_n$. The *Algebra Theorem* shows that the elbow stopping rule achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without oracle knowledge of the mixing rate, lag, or smoothness index. The *Dynamics Theorem* shows that under uniform separation, empirical delay vectors constitute a geometric witness converging at rate $n^{-\beta/(2\beta+d)}$. The *Conjunction Theorem* shows that for deterministic T , the algebraic and geometric witnesses certify the same event — not merely correlated, but identical.

All three witnesses, together with a fourth information-theoretic one from the collision entropy of the delay-vector distribution, measure the same failure: separation collapsing across delay fibres. Under fibre mixing, they are equivalent. The convergence of three independently motivated certifications onto a single geometric object is the main message.

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1 Introduction

A probability measure on an observable algebra already contains, without further assumption, a conditional regularity structure — the systematic, outcome-by-outcome account of how one observation is distributed given another [27]. Concretely, the Rokhlin disintegration of any two observations Q, F gives a Markov kernel $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$ that is forced by the measure, not chosen. Composing with Q gives the minimal sufficient factor Q_* , the coarsest reduction of the sample point carrying full conditional information about F . In a measure-preserving system with $Q_t = h \circ T^t$, this factor becomes the delay map Φ_h , and the reconstruction question becomes internal: is the factor faithful, or does it see only a proper quotient of X ? The reconstruction theorem of [27] answers in terms of an observable algebra condition — $\mathcal{O}_h = \sigma\{h \circ T^n : n \in \mathbb{Z}\}$ generates $\mathcal{B}$ modulo μ if and only if Φ_h is a measure-theoretic embedding. This is an exact asymptotic result; it says nothing about rates, certificates, or what can be inferred from a finite time series. The results of [28, 27] are taken as given; this paper asks what finite data can certify about the structure they establish.

The certification question has two natural levels, and the paper works from the more general inward. At the level of a general query system [28] — before any dynamical or geometric structure is assumed — one can ask: given finite responses to a directed family of queries, does

the empirical evidence certify that the observable structure is resolving correctly, and can CE-failure be detected from data? Section 2 answers this at full generality. The central objects are the population separation defect $\delta(\mathcal{G})$ and its empirical U-statistic proxy; the main results are a population convergence theorem (CE drives the defect to zero under honest refinement), an estimation theorem (the empirical proxy concentrates under IID sampling), and a CE-shadow corollary (a persistent empirical floor is evidence against CE). The corollary has no temporal hypothesis; the admissibility condition CE acquires an empirical signature that is entirely independent of dynamics.

Moving to the dynamical specialisation, the two threads of the programme converge. Section 2's certification framework meets the reconstruction theorem of [27], and the meeting point is stronger than either alone. In the general query setting, honest refinement is a design condition: it must be built into the query family. In the dynamical setting, it is automatic — a consequence of the reconstruction theorem itself, which guarantees that $\sigma(\bigcup_L \mathcal{O}_h^{(L)}) = \mathcal{B}$ modulo μ once the factor is faithful. The observer does not need to design an honest query path; the dynamics provide it. The remaining degrees of freedom — mixing rate and embedding dimension — are what the rate theorems exploit. The defect specialises to $\delta(L) = \delta(\mathcal{O}_h^{(L)})$. Three theorems certify reconstruction in this regime.

The Algebra Theorem (Section 6). The elbow stopping rule $\hat{L}^*$ fires when the empirical defect $\hat{\delta}(L, n)$ stops decreasing. Under exponential mixing with resolvability constant $C_\lambda > 4$, $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d -dimensional state space, without any oracle knowledge of the mixing rate, the lag $L^*(n)$, or the smoothness index s .

The Dynamics Theorem (Section 7). When $T \in C^r$ and the uniform separation condition (US) holds, the delay map $\Phi_h^{(L)}$ is bi-Lipschitz, and the empirical delay vectors constitute a second witness certifying that points are being separated by the dynamics. This kernel witness converges at rate $n^{-\beta/(2\beta+d)}$ and fails precisely when (US) fails.

The Conjunction Theorem (Section 8). For deterministic T , the TV separation $d_L(x, x') = 1$ if and only if x and x' are separated by $\mathcal{O}_h^{(L)}$ — the algebraic identity $\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$ is the Markov bridge. Under reconstruction and (US), both witnesses certify the same object; the theorem names two failure modes (reconstruction failure collapses the algebra witness; (US)-failure collapses the kernel witness) that jointly exhaust the ways certification can fail.

The discussion section records where each proof requires mixing and what would be needed to extend beyond it.

2 General observational certification

Fix a query system $(\mathcal{E}_n)_n$ with observable σ -algebra $\sigma(\mathcal{C})$ and realization measure μ satisfying CE [28]. For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$, the *population separation defect* is

$$\delta(\mathcal{G}) := \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E).$$

It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C}) \bmod \mu$. A scheme $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Theorem 2.1 (Population theorem). *Under CE and any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \longrightarrow 0.$$

Proof. Since $(\mathcal{G}_k)$ is honest, every $A \in \sigma(\mathcal{C})$ is $\sigma(\bigcup_k \mathcal{G}_k)$ -measurable, so the martingale $\mathbb{E}_\mu[\mathbf{1}_A | \mathcal{G}_k]$ converges to $\mathbf{1}_A$ in $L^1(\mu)$. Thus $\inf_{E \in \mathcal{G}_k} \mu(A \triangle E) \rightarrow 0$ for each A . CE [28] ensures this convergence is uniform: no mass persists on the residual sets. Hence $\delta(\mathcal{G}_k) \rightarrow 0$. $\square$

Given N independent query responses $\omega_1, \dots, \omega_N$ drawn from μ , let $R_i = (Q_{n_1}(\omega_i), \dots, Q_{n_k}(\omega_i)) \in \alpha_{n_1} \times \dots \times \alpha_{n_k}$ be the joint response vector of the i -th sample. Define the *empirical separation defect*

$$\hat{\delta}(\mathcal{G}, N) := \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \mathbf{1}[R_i = R_j],$$

the empirical collision probability of the joint response. This is a second-order U-statistic requiring no geometric structure on Ω .

Theorem 2.2 (Estimation theorem). *Under IID sampling from the observable law of the joined query responses, $\hat{\delta}(\mathcal{G}, N) \rightarrow \delta(\mathcal{G})$ in probability as $N \rightarrow \infty$.*

Proof. $\hat{\delta}(\mathcal{G}, N)$ is a second-order U-statistic for a bounded kernel; the result is an instance of the standard U-statistic law of large numbers [16]. $\square$

Corollary 2.3 (CE-shadow). *Under an honest refinement scheme and IID sampling, if $\hat{\delta}(\mathcal{G}_k, N) \geq c > 0$ for all large k and N , that is evidence against CE: CE is what guarantees the floor can be discharged by further refinement.*

Remark 2.4 (Scope). The corollary is one-directional: persistent floor implies CE-failure; $\delta(\mathcal{G}_k) \rightarrow 0$ does not imply CE. The defect is defined inside $\sigma(\mathcal{C})$, while CE also controls escape to the Stone boundary — non-principal ultrafilters invisible to the defect functional. A charge family can resolve $\sigma(\mathcal{C})$ completely while still placing mass on non-realised boundary points [28].

Honest refinement is essential: a stalling empirical defect under a poorly-chosen query path is a design artifact, not an admissibility failure. Under an honest scheme it is the latter.

CE thereby acquires an empirical signature absent from its abstract treatment in [28]: it is not only the admissibility condition for σ -additivity (Theorem I:thm:sp1 of [28]), but a structural hypothesis with a detectable finite-sample consequence.

Remark 2.5 (Non-smooth instance). The general framework of this section applies without any smoothness, metric, or manifold hypothesis. As an illustration, let $\mathcal{C}$ be the middle-thirds Cantor set with its shift-invariant Cantor measure $\mu_{\mathcal{C}}$ and shift map S , as in [27]. The cylinder partition $\mathcal{G}_L = \mathcal{O}_h^{(L)}$ — the algebra generated by the first $L+1$ delay coordinates — is an honest refinement scheme: it is increasing and $\sigma(\bigcup_L \mathcal{G}_L) = \mathcal{B}(\mathcal{C}) \bmod \mu_{\mathcal{C}}$ by the reconstruction theorem. Each fibre has $\mu_{\mathcal{C}}$ -measure $2^{-(L+1)}$, so the separation defect satisfies $\delta(\mathcal{G}_L) \leq 2^{-(L+1)}$: geometric decay in L , with no appeal to Riemannian geometry, Hölder classes, or mixing rates. The CE-shadow corollary applies directly — a persistent empirical collision floor is evidence against CE — and the information horizon $L^* \approx \log_2 N$ marks where the empirical defect estimate becomes unreliable, for exactly the reason noted in Remark 3.1. The smooth dynamical specialisation that follows is one regime; this is another.

The remainder of this paper establishes what becomes achievable when the general framework of this section meets the dynamical structure of [27]. Specialisation to a dynamical setting might seem to narrow the question; instead it strengthens it. Honest refinement, which must here be a design condition, becomes free; the separation defect, which is abstract, becomes geometrically interpretable; and the witnesses, which here are qualitative (floor or no floor), become quantitative with optimal rates. That gain is the subject of Sections 6–8.

3 Setup and Notation

The measure-preserving system. Let $(X, \mathcal{B}, \mu)$ be a probability space where X is a compact smooth d -dimensional Riemannian manifold and μ is a Borel probability measure with smooth, everywhere-positive density $\rho : X \rightarrow \mathbb{R}_{>0}$ satisfying

$$\rho_{\min} := \operatorname{ess\,inf}_X \rho > 0.$$

Let $T : X \rightarrow X$ be a μ -preserving measurable map. We make no injectivity or invertibility assumption on T beyond what is stated in each hypothesis.

The observable and the delay map. Fix an observable $h \in L^\infty(\mu)$ with $\|h\|_{L^\infty} =: a < \infty$. For each lag depth $L \geq 0$ define the *delay map*

$$\Phi_h^{(L)} : X \rightarrow \mathbb{R}^{L+1}, \quad \Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x)).$$

The *observable σ -algebra at lag L* is

$$\mathcal{O}_h^{(L)} := \sigma(\Phi_h^{(L)}) = \sigma\{h \circ T^k : 0 \leq k \leq L\}.$$

The *delay-polynomial class* of degree $D \geq 1$ is

$$\mathcal{A}_h^{(L)} := \{p \circ \Phi_h^{(L)} : p \text{ is a polynomial of degree } \leq D \text{ on } \mathbb{R}^{L+1}\} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu).$$

Since $\Phi_h^{(L)}(X) \subseteq [-a, a]^{L+1}$, all elements of $\mathcal{A}_h^{(L)}$ are bounded.

The approximation error. The central quantity is the σ -algebra approximation error

$$\delta(L) := \delta(\mathcal{O}_h^{(L)}) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E), \quad (1)$$

which measures how much of $\mathcal{B}$ remains outside $\mathcal{O}_h^{(L)}$. It is non-increasing in L , vanishes if and only if reconstruction holds, and under (G1)–(G2) on a smooth manifold reaches zero by lag $L_0 \leq d - 1$ [27].

Fibre structure and collision probability. The delay map $\Phi_h^{(L)}$ partitions X into *fibres*: for each $z \in \mathbb{R}^{L+1}$, define

$$F_z := (\Phi_h^{(L)})^{-1}(z), \quad p_z := \mu(F_z), \quad \mu_z := \mu(\cdot \mid F_z).$$

Call z an ε -large fibre if $p_z \geq \varepsilon$. The push-forward $\nu_L := (\Phi_h^{(L)})_* \mu$ is the distribution of delay vectors.

The set of unseparated pairs $R_L := \{(x, x') : \Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')\}$ decomposes as $R_L = \bigsqcup_z F_z \times F_z$, giving

$$(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z). \quad (2)$$

This quantity is the *collision probability* of the delay-vector distribution: it is the probability that two independent draws from μ produce the same delay vector.

Remark 3.1 (Information horizon). The fibre structure imposes a hard constraint on what finite data can certify. At lag L the observable algebra partitions X into at most 2^{L+1} atoms (in the binary-coded case; more generally, the number of fibres grows exponentially with L). With a sample of size n , the mean occupancy per fibre is $n/2^{L+1}$, which halves with each additional lag. The *information horizon* is

$$L^* := \lfloor \log_2 n \rfloor,$$

the lag at which mean fibre occupancy drops to 1. Beyond L^* the sample is partitioned across more fibres than it contains points, so empirical fibre statistics become unreliable. This gives a first-principles derivation — from the observable algebra structure alone — of why the concentration rate $\varepsilon_n \sim n^{-2s/(2s+d)}$ and the elbow stopping rule of Theorem 8.3 fire when they do: doubling the sample size buys exactly one more trustworthy lag.

The data model. We observe a time series $x_1, x_2, \dots, x_n$ which is an orbit segment of T : $x_i = T^{i-1}x_1$ for some initial condition $x_1 \in X$ drawn from μ . The empirical measure is $\mu_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. We do not assume ergodicity or density of the orbit; each result states the mixing hypothesis it uses.

Structural hypotheses. The following hypotheses are referenced throughout. They are not assumed globally; each theorem states exactly which subset it uses.

- (G1) *No short periodic orbits:* T has no periodic orbit of period $\leq L$ in $\text{supp}(\mu)$. This ensures the orbit points $x, Tx, \dots, T^L x$ are distinct μ -a.e., which is used in the Sard–Smale argument for full rank of $\Phi_h^{(L)}$.
- (G2) *Non-degenerate observable:* $h \in C^2(X)$ with $dh \neq 0$ μ -a.e. This places h in the Sard–Smale regular-value residual set and ensures $\Phi_h^{(L)}$ has rank $\min(L+1, d)$ generically.
- (US) *Uniform separation:* $\inf_{x \in X} \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, where $\sigma_{\min}$ denotes the smallest singular value. This is a uniform lower Lipschitz bound on $\Phi_h^{(L)}$, strictly stronger than (G1)–(G2) which give full rank only μ -a.e. Condition (US) is used only in the Dynamics and Conjunction theorems (Sections 7 and 8).

The empirical witness. The empirical analogue of $\delta(L)$ is

$$\hat{\delta}(L, n) := \sup_{p \in \mathcal{P}_D^{(L)}, \|p\|_{L^2(\mu)} \leq 1} \|p - \hat{\mathbb{E}}[p | \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2,$$

where $\mathcal{P}_D^{(L)}$ is the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$ (normalised in $L^2(\mu)$), and $\hat{\mathbb{E}}[p | \Phi_h^{(L)}]$ is the Nadaraya–Watson kernel regression estimator (Definition 5.5).

The main result of Section 5 is that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$ at rate $\varepsilon_n = C n^{-2s/(2s+d)}$ under exponential mixing, provided $d > 2s$.

4 Bias Bound and Conditional Variance

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub- σ -algebra, with $\delta(\mathfrak{m})$ as in (1).

Lemma 4.1 (Conditional variance identity). *For any $S \in \mathcal{B}$:*

$$\mathbb{E}[\text{Var}(\mathbb{1}_S | \mathcal{O}_h^{(L)})] = \frac{1}{2} \int_{R_L} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 d(\mu \otimes \mu)(x, x').$$

Proof. By disintegration, $(\mu \otimes \mu)|_{R_L} = \int \mu_z \otimes \mu_z d\nu_L(z)$. On each fibre F_z :

$$\frac{1}{2} \int_{F_z \times F_z} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 d(\mu_z \otimes \mu_z) = \mu_z(S) \mu_z(S^c).$$

Integrating with weight p_z : $\frac{1}{2} \int_{R_L} |\mathbb{1}_S - \mathbb{1}_{S'}|^2 d(\mu \otimes \mu) = \int p_z \mu_z(S) \mu_z(S^c) d\nu_L(z)$. Since $\text{Var}(\mathbb{1}_S | \mathcal{O}_h^{(L)})(x) = \mu_z(S) \mu_z(S^c)$ on F_z , both sides equal $\int \mu_z(S) \mu_z(S^c) d\nu_L(z)$. $\square$

Bounding $|\mathbb{1}_S - \mathbb{1}_{S'}|^2 \leq 1$ in the lemma and taking the supremum gives $\delta(L) \leq \frac{1}{2}(\mu \otimes \mu)(R_L)$.

Remark 4.2 (Finite analogue of collective exhaustion). This bound and its converse (Corollary 6.12, proved under fibre mixing in §6) together say: algebraic indistinguishability $\delta(L)$ vanishes if and only if the mass of indistinguishable pairs $(\mu \otimes \mu)(R_L)$ vanishes. This is the finite-sample counterpart of collective exhaustion [28]. There, CE asserts that set-theoretic vanishing must be witnessed at some finite level of the refinement order. Here, the analogous

principle is that observational indistinguishability cannot persist globally without manifesting as a non-vanishing mass of indistinguishable pairs at finite lag L . The fibre mixing condition plays an analogous structural role to CE: it is a sufficient condition under which the global and pairwise witnesses are comparable. Whether fibre mixing is itself derivable from dynamical conditions — or is genuinely irreducible, as CE is proved to be in [28] — remains open; see the open problems in Section 9.

Lemma 4.3 (Bias bound). *For every $f \in L^\infty(\mu)$,*

$$\|f - \mathbb{E}[f \mid \mathbf{m}]\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(\mathbf{m})^{1/2}.$$

Proof. For an indicator $\mathbb{1}_S$: the conditional expectation is the L^2 -projection, so $\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathbf{m}]\|_{L^2}^2 \leq \mu(S \triangle E) \leq \delta(\mathbf{m})$ for the best $E \in \mathbf{m}$. The general bounded case follows by the layer-cake representation $f = \int_{-M}^M \mathbb{1}_{\{f > t\}} dt$ and Minkowski's inequality for integrals, giving $\|f - \mathbb{E}[f \mid \mathbf{m}]\|_{L^2} \leq 2M \cdot \delta(\mathbf{m})^{1/2}$. $\square$

In the delay setting of Section 3, let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 4.4 (Three-term decomposition). *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{\text{bias}} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{\text{approx.}} + \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{\text{statistical}}, \quad (3)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (1) and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 4.3. $\square$

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n .

Lemma 4.5 (Observable reverse lower bound). *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 as $n \rightarrow \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in (1) at lag L and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ is the empirical measure.

Proof. *Step A (Population CE gap).* Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \leq \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 = \mu(S^* \triangle E) \geq \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \rightarrow 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2 -projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n , $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C'n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

5 Concentration of the Empirical Witness

Throughout this section: (T, h) satisfies (G1)–(G2), X is a compact smooth d -manifold, μ has smooth positive density, and $d > 2s$.

5.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \leq 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 5.1 (Uniform boundedness). *Define*

$$M_{D,L} := \left(\frac{D+L+1}{L+1} \right)^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \text{essinf}_X \rho > 0$ is the infimum of the density of μ . Then for every $p \in \mathcal{P}_D^{(L)}$:

$$\|p\|_{L^\infty} \leq M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to $y = x/a \in [-1, 1]^{L+1}$ and expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \leq D}$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives $|\tilde{p}(y)| \leq N(D, L+1)^{1/2} (\sum |c_\alpha|^2)^{1/2}$ with $N(D, L+1) = \binom{D+L+1}{L+1}$. The Legendre orthogonality $\|P_\alpha\|_{L^2}^2 = \prod_i (2\alpha_i + 1)^{-1}$ gives $\sum |c_\alpha|^2 \leq (2D+1) \|\tilde{p}\|_{L^2}^2$ (the maximum $(2D+1)$ is L -independent, achieved at $\alpha = (D, 0, \dots, 0)$). Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\text{vol})}$ and $\|p\|_{L^2(\text{vol})} \leq \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. $\square$

5.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} = \left\{ x \mapsto (p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x))^2 : p \in \mathcal{P}_D^{(L)}, \|p\|_{L^2(\mu)} \leq 1 \right\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 5.5). Then $\hat{\delta}(L, n) = \sup_{f \in \mathcal{F}_{L,D}} \frac{1}{n} \sum_{i=1}^n f(x_i)$.

Proposition 5.2 (Concentration, $\star$). *Assume $d > 2s$. There exist constants $c, C > 0$ depending only on $(d, s, \|h\|_{L^\infty})$ such that for all $t > 0$ and all $n \geq 1$:*

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t) \leq 2 \exp\left(-\frac{c n t^2}{M_{D,L}^4 \cdot D^d}\right). \quad (\star)$$

In particular, setting $t = \varepsilon_n/2$ with $\varepsilon_n = C' n^{-2s/(2s+d)}$:

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > \varepsilon_n/2) \leq 2 \exp\left(-c' n^{(d-2s)/(2s+d)}\right) \rightarrow 0.$$

Proof. Step 1 (Boundedness). By Lemma 5.1, every $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$ satisfies $\|p\|_{L^\infty} \leq M_{D,L}$. Hence every $f = (p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}])^2 \in \mathcal{F}_{L,D}$ satisfies $\|f\|_{L^\infty} \leq 4M_{D,L}^2 =: B_{L,D}^2$.

Step 2 (McDiarmid). Replacing one observation x_i with x'_i changes $\hat{\delta}(L, n)$ by at most $B_{L,D}^2/n$. By the McDiarmid bounded-difference inequality [29]:

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t) \leq 2 \exp\left(-\frac{2n^2 t^2}{n \cdot (B_{L,D}^2/n)^2}\right) = 2 \exp\left(-\frac{n t^2}{2B_{L,D}^4/n}\right).$$

Step 3 (Rademacher bias). By the symmetrisation inequality [3]:

$$\mathbb{E}[\hat{\delta}(L, n)] - \delta(L) \leq 2 \mathfrak{R}_n(\mathcal{F}_{L,D}).$$

By the Ledoux–Talagrand contraction lemma [22], since $f = (p - \cdot)^2$ is $2B_{L,D}$ -Lipschitz in the second argument:

$$\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq 2B_{L,D} \mathfrak{R}_n(\mathcal{P}_D^{(L)}).$$

In the Takens regime the image $\Phi_h^{(L)}(X)$ is a d -dimensional submanifold, so the VC dimension of $\mathcal{P}_D^{(L)}$ restricted to $\Phi_h^{(L)}(X)$ is $O(D^d)$ [12]. Hence $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \leq C\sqrt{D^d/n}$, giving $\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq C M_{D,L} \sqrt{D^d/n}$.

Step 4 (Combined). Combining Steps 2–3 and absorbing constants:

$$P(|\hat{\delta}(L, n) - \delta(L)| > t) \leq 2 \exp\left(-\frac{c n t^2}{M_{D,L}^4 D^d}\right).$$

Setting $t = \varepsilon_n/2 = C' n^{-2s/(2s+d)}/2$, the exponent becomes

$$c n \varepsilon_n^2 / (M_{D,L}^4 D^d) \propto n^{1-4s/(2s+d)} = n^{(d-2s)/(2s+d)}.$$

This tends to $+\infty$ if and only if $d > 2s$, proving the final claim. $\square$

Remark 5.3 (The $d > 2s$ hypothesis). When $d \leq 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [2] is technically available but left to future work; we take $d > 2s$ as a standing hypothesis.

5.3 Source 1: the LLN gap

Lemma 5.4 (LLN gap). *With notation as above:*

$$\sup_{p \in \mathcal{P}_D^{(L)}} \left| \|p - \mathbb{E}[p | \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2 - \|p - \mathbb{E}[p | \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \right| \leq C \sqrt{D^d/n}$$

almost surely as $n \rightarrow \infty$, where C depends on $(d, D, M_{D,L})$.

Proof. The class $\{(p - \mathbb{E}[p | \Phi_h^{(L)}])^2 : p \in \mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [12] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. $\square$

5.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p | \Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 5.5 (Nadaraya–Watson estimator). For bandwidth $h_n > 0$ and kernel $K : \mathbb{R}^{L+1} \rightarrow \mathbb{R}_{\geq 0}$ (compactly supported, symmetric, $\int K = 1$):

$$\hat{\mathbb{E}}[p | \Phi_h^{(L)}(x)] := \frac{\sum_{j=1}^n K\left(\frac{\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x_j)}{h_n}\right) p(x_j)}{\sum_{j=1}^n K\left(\frac{\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x_j)}{h_n}\right)}.$$

Proposition 5.6 (Source 2 estimator bias). *Assume:*

- (i) $T, h \in C^r$ with $r \geq 2$, μ smooth positive density ρ with $\rho_{\min} > 0$.
- (ii) $\Phi_h^{(L)}$ has full rank μ -a.e. (follows from (G1)–(G2) and Lemma 6.2).

(iii) T is exponentially mixing with rate $\lambda > 0$ and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

(iv) The NW bandwidth satisfies $h_n \sim n^{-1/(2\beta+d_{\text{eff}}(L))}$ with $d_{\text{eff}}(L) = \min(L+1, d)$ and $\beta \geq 1$.

Then there exists $C_2(D, L) > 0$ such that for all $p \in \mathcal{P}_D^{(L)}$:

$$\mathbb{E} \|\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \leq C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\text{eff}}(L))},$$

uniformly in $p \in \mathcal{P}_D^{(L)}$, where $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$ and A is a constant depending on $(\beta, d_{\text{eff}}(L), \rho_{\min})$.

Proof. The proof is a four-link chain.

Link 1 (Smooth disintegration). Under conditions (i)–(ii), the Rokhlin disintegration $\mu = \int \mu_z d((\Phi_h^{(L)})_* \mu)(z)$ has fibre measures μ_z supported on the smooth fibres $F_z = \{\Phi_h^{(L)} = z\}$. An implicit-function-theorem argument (three sub-steps: transfer to common surface via a C^{r-1} diffeomorphism; bound weight variation; bound Jacobian factor) gives:

$$\|\mu_z - \mu_{z'}\|_{\text{TV}} \leq C_1 |z - z'|^\beta, \quad \beta = 1 \text{ (Lipschitz floor)},$$

where C_1 depends on $(\|T\|_{C^r}, \|h\|_{C^r}, \rho_{\min})$. (The optimal exponent $\beta = r - 1$ follows from the full Taylor expansion of the fibre weight; the floor $\beta = 1$ is sufficient for convergence.)

Link 2 (Bernstein–Markov). From Link 1 and Lemma 5.1: for $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$,

$$|\mathbb{E}[p \mid \Phi_h^{(L)} = z] - \mathbb{E}[p \mid \Phi_h^{(L)} = z']| = \left| \int p d\mu_z - \int p d\mu_{z'} \right| \leq M_{D,L} \cdot C_1 \cdot |z - z'|^\beta.$$

Hence $z \mapsto \mathbb{E}[p \mid \Phi_h^{(L)} = z]$ is Hölder(β) with constant $\Lambda_{D,L} := M_{D,L} C_1$, uniformly over $p \in \mathcal{P}_D^{(L)}$.

Link 3 (NW bias-variance balance). Standard NW theory [12] for a Hölder(β) regression function on a $d_{\text{eff}}(L)$ -dimensional domain gives:

$$\mathbb{E} \|\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \leq \Lambda_{D,L}^2 h_n^{2\beta} + \frac{C_K M_{D,L}^2}{n h_n^{d_{\text{eff}}(L)} f_{\min}},$$

where C_K depends on the kernel K and $f_{\min}$ is the minimum of the marginal density of $(\Phi_h^{(L)})_* \mu$. The $M_{D,L}^2$ factor appears in both terms. Balancing: $\Lambda_{D,L}^2 h_n^{2\beta} = C_K M_{D,L}^2 / (n h_n^{d_{\text{eff}}(L)} f_{\min})$ gives $h_n^* \sim A \cdot n^{-1/(2\beta+d_{\text{eff}}(L))}$ where A depends only on $(\beta, d_{\text{eff}}(L), C_K, f_{\min}, C_1)$ — not on D, L , or p . The $M_{D,L}^2$ factors cancel from both sides of the balance equation.

Link 4 (Uniformity). Substituting h_n^* , both terms contribute $C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\text{eff}}(L))}$ with $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$. Since the bound is p -free, it holds uniformly over $\mathcal{P}_D^{(L)}$. $\square$

Theorem 5.7 (Full convergence $\hat{\delta}(L, n) \rightarrow \delta(L)$). *Under the hypotheses of Proposition 5.6 and with $d > 2s$:*

$$\mathbb{E} |\hat{\delta}(L, n) - \delta(L)| \leq C \sqrt{D^d/n} + C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\text{eff}}(L))}$$

and $P(|\hat{\delta}(L, n) - \delta(L)| > \varepsilon_n) \rightarrow 0$ exponentially, where $\varepsilon_n = C' n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L, n) - \delta(L)| \leq \underbrace{|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]|}_{\text{concentration, } (\star)} + \underbrace{|\mathbb{E}[\hat{\delta}(L, n)] - \delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C \sqrt{D^d/n}$ (Lemma 5.4); Source 2 is bounded by $C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\text{eff}}(L))}$ (Proposition 5.6). The concentration term is bounded by $(\star)$ (Proposition 5.2). Combining at $t = \varepsilon_n/2$ gives the probability statement. $\square$

6 The Algebra Theorem

This section assembles the results of Sections 4 and 5 into the Algebra Theorem: the data-driven stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without knowing the oracle lag $L^*(n)$ or the mixing rate of T .

6.1 Effective dimension and the piecewise rate

Definition 6.1 (Effective dimension). The *effective dimension at lag L* is

$$d_{\text{eff}}(L) := \text{rank}(D\Phi_h^{(L)}|_x) \quad \text{at a generic point } x \in X.$$

(The rank is constant on an open dense set by the rank theorem, so this is well-defined μ -a.e.)

Lemma 6.2 (d_{eff} step function). *Under (G1)–(G2) with X a compact smooth d -manifold and μ smooth positive density:*

$$d_{\text{eff}}(L) = \min(L + 1, d) \quad \text{for all } L \geq 0.$$

Proof. The upper bound $d_{\text{eff}}(L) \leq \min(L + 1, d)$ holds because $D\Phi_h^{(L)}|_x$ is a linear map from $T_x X$ (dimension d) to $\mathbb{R}^{L+1}$.

For the lower bound we use the Sard–Smale theorem [34] applied to two failure maps.

Case $L + 1 \leq d$. Rank $< L + 1$ at x means the cotangent vectors $\omega_j(x) := (DT^j|_x)^*(dh|_{T^j x})$ are linearly dependent. Define the failure map

$$\Psi : C^2(X) \times X \times \mathbb{P}^L \rightarrow T^*X, \quad (h, x, [\lambda]) \mapsto d\left(\sum_j \lambda_j h \circ T^j\right)|_x.$$

By (G1), the orbit points $T^j x$ are distinct μ -a.e., so bump functions at $T^j x$ can prescribe any cotangent vector at x independently. Hence $D_h \Psi$ is surjective onto $T_x^* X$ (dimension d), and $\Psi^{-1}(0)$ is a Banach submanifold of codimension d . The projection to $C^2(X)$ is Fredholm of index $d + L - d = L$, so by Sard–Smale the regular values form a residual set $R \subset C^2(X)$. For $h \in R$, the failure locus in X has dimension $\leq L < d$, hence μ -measure zero.

By (G2), $dh \neq 0$ μ -a.e., which forces every critical point of $F_\lambda = \sum_j \lambda_j h \circ T^j$ to lie in the codimension- d submanifold $S_\lambda = \{dh_x \in \text{span}\{\omega_1(x), \dots, \omega_L(x)\}\}$ (a finite set for each $[\lambda]$, since X is d -dimensional). Hence h is a regular value of π , i.e., $h \in R$.

Case $L + 1 > d$. Immersion failure requires $\exists v \neq 0$ with $D\Phi_h^{(L)}|_x \cdot v = 0$. Define

$$\Xi : C^2(X) \times (TX \setminus 0) \rightarrow \mathbb{R}^{L+1}, \quad (h, (x, v)) \mapsto (dh_{T^j x}(DT^j|_x \cdot v))_{j=0}^L.$$

By (G1), $D_h \Xi$ is surjective onto $\mathbb{R}^{L+1}$, so $\Xi^{-1}(0)$ has codimension $L + 1$ and the projection to $C^2(X)$ is Fredholm of index $2d - 1 - (L + 1) = 2d - L - 2$. For $L \geq d$: $2d - L - 2 \leq d - 2 < d - 1$, so the failure locus in X has dimension $< d - 1$ and μ -measure zero. Condition (G2) places h in the regular-value set by the same argument as Case 1. $\square$

Proposition 6.3 (Piecewise pre-Takens rate). *Assume (G1)–(G2), X compact smooth d -manifold, μ smooth positive density, $f \in \text{H\"older}(s) \cap L^\infty(\mu)$, $d > 2s$, and $T \in C^r$ with $r \geq 2$. Let $\hat{f}_n^{(L)}$ be the empirical risk minimiser over $\mathcal{A}_h^{(L)}$ with $D = D^*(n, L)$ chosen to balance approximation and statistical terms. Then for each $L \geq 0$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d_{\text{eff}}(L))}$$

with probability tending to 1, where $d_{\text{eff}}(L) = \min(L + 1, d)$. In the Takens regime ($L \geq d - 1$): $d_{\text{eff}}(L) = d$ and the rate locks at $n^{-s/(2s+d)}$.

Proof. Apply the three-term decomposition (Lemma 4.4). The bias term is bounded by $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}$ (Lemma 4.3). By Lemma 6.2, $\Phi_h^{(L)}(X)$ is a $d_{\text{eff}}(L)$ -dimensional smooth submanifold. On this submanifold, $\mathbb{E}[f \mid \mathcal{O}_h^{(L)}](x) = g^*(\Phi_h^{(L)}(x))$ for some $g^* \in \text{Hölder}(s)$; standard polynomial approximation on a $d_{\text{eff}}(L)$ -dimensional smooth domain gives approximation error $\leq C_{\text{approx}} \cdot D^{-s/d_{\text{eff}}(L)}$. The statistical term is $\leq C_{\text{stat}}(D^{d_{\text{eff}}(L)}/n)^{1/2}$ by Proposition 5.2 with d replaced by $d_{\text{eff}}(L)$. Balancing: $D^* \sim n^{d_{\text{eff}}(L)/(2s+d_{\text{eff}}(L))}$ and both terms are $O(n^{-s/(2s+d_{\text{eff}}(L))})$. $\square$

6.2 The oracle lag and the elbow stopping rule

Definition 6.4 (Oracle lag). The *oracle lag* is

$$L^*(n) := \min\{L \geq 0 : \delta(L) \leq C_0 \cdot n^{-2s/(2s+d)}\},$$

where $C_0 > 0$ is a constant depending on $\|f\|_{L^\infty}$. At $L = L^*(n)$, the bias term $2\|f\|_{L^\infty} \cdot \delta(L^*)^{1/2}$ is $O(n^{-s/(2s+d)})$, matching the statistical floor.

The oracle lag $L^*(n)$ requires knowledge of the mixing rate, which is unobservable. The empirical witness $\hat{\delta}(L, n)$ sidesteps this: it measures reconstruction quality directly from data.

Definition 6.5 (Elbow stopping rule). Fix a tolerance sequence $\tau_n > 0$. The *elbow stopping rule* is

$$\hat{L}^* := \min\{L \geq 0 : \hat{\delta}(L+1, n) - \hat{\delta}(L, n) > -\tau_n\}.$$

Theorem 6.6 (Elbow location). Assume the hypotheses of Proposition 6.3 and exponential mixing $\delta(L) \leq Ae^{-\lambda L}$ with resolvability constant $C_\lambda := (1 - e^{-\lambda})Ae^\lambda/C' > 4$. Set $\tau_n = \frac{C_\lambda}{2} \cdot C' \cdot n^{-2s/(2s+d)}$. Then

$$P(\hat{L}^* = L^*(n)) \rightarrow 1 \quad \text{exponentially in } n.$$

Proof. We show the stopping criterion fires at $L = L^*(n)$ (no overshoot) and not before (no early stop).

No overshoot: $\hat{L}^* \leq L^*(n)$. At $L = L^*(n)$, the population increment satisfies $|\delta(L^*(n) + 1) - \delta(L^*(n))| \leq \delta(L^*(n)) \leq C_0 n^{-2s/(2s+d)}$ (non-increasing δ). By Theorem 5.7, $|\hat{\delta}(L, n) - \delta(L)| \leq \varepsilon_n = C' n^{-2s/(2s+d)}$ whp. The empirical increment satisfies $\hat{\delta}(L^* + 1, n) - \hat{\delta}(L^*, n) \geq -C_0 n^{-2s/(2s+d)} - 2\varepsilon_n > -\tau_n$ whp, so the criterion fires.

No early stop: $\hat{L}^* \geq L^*(n)$. For $L < L^*(n)$, exponential mixing gives population decrement $\delta(L) - \delta(L+1) \geq \Delta(L) \geq C_\lambda \cdot C' n^{-2s/(2s+d)}$. Tracking via Theorem 5.7:

$$\hat{\delta}(L, n) - \hat{\delta}(L+1, n) \geq C_\lambda C' n^{-2s/(2s+d)} - 2\varepsilon_n = (C_\lambda - 2)C' n^{-2s/(2s+d)}.$$

Since $C_\lambda > 4$: $(C_\lambda - 2)C' > \tau_n \cdot 2/C_\lambda \cdot C_\lambda = 2\tau_n/C_\lambda \cdot C_\lambda$, and specifically $(C_\lambda - 2)C' n^{-2s/(2s+d)} \geq \tau_n$ when $C_\lambda - 2 \geq C_\lambda/2$, i.e., $C_\lambda \geq 4$. So the stopping criterion does *not* fire for $L < L^*(n)$, and $\hat{L}^* \geq L^*(n)$.

Remark 6.7 (Scope of the elbow theorem). The resolvability condition $C_\lambda > 4$ is not an artefact: when $C_\lambda \leq 4$ the elbow drop at $L^*(n)$ merges with the noise floor ε_n and the two are indistinguishable from data. For polynomial mixing, $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \rightarrow 0$ relative to ε_n , so the elbow drops vanish asymptotically and the theorem fails. A separate argument (e.g. integrated elbow) is needed for that case and is not proved here. $\square$

Theorem 6.8 (Algebra Theorem). Assume:

- (i) Hypotheses of Proposition 6.3: (G1)–(G2), X compact smooth d -manifold, μ smooth positive density, $f \in \text{Hölder}(s) \cap L^\infty(\mu)$, $d > 2s$, $T \in C^r$, $r \geq 2$.

(ii) *Exponential mixing*: $\delta(L) \leq Ae^{-\lambda L}$ for some $A, \lambda > 0$.

(iii) *Resolvability*: $C_\lambda > 4$.

(iv) *Exponential mixing for the estimator bias* (Proposition 5.6) and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

Let $\hat{L}^*$ be the elbow stopping rule (Definition 6.5) with $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$. Then

$$\|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} = O_p(n^{-s/(2s+d)}).$$

The rate $n^{-s/(2s+d)}$ is minimax-optimal for Hölder(s) functions on a d -dimensional domain [35]. No oracle input (mixing rate, $L^*(n)$, Sobolev index s) is required.

Proof. On the event $\{\hat{L}^* = L^*(n)\}$, which has probability tending to 1 by Theorem 6.6:

$$\begin{aligned} \|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} &= \|f - \hat{f}_n^{(L^*(n))}\|_{L^2(\mu)} \\ &\leq 2\|f\|_{L^\infty} \cdot \delta(L^*(n))^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d)} \\ &\leq 2\|f\|_{L^\infty} \cdot (C_0 n^{-2s/(2s+d)})^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d)} \\ &= O(n^{-s/(2s+d)}). \end{aligned} \quad \square$$

6.3 Entropy characterisation of reconstruction

Definition 6.9 (Fibre mixing). Fix $\varepsilon > 0$. An event $S^* \in \mathcal{B}$ is (c, ε) -mixing across fibres if there exists $c > 0$ such that

$$\mu_z(S^*) \mu_z((S^*)^c) \geq c \cdot p_z \quad \text{for } \nu_L\text{-a.e. } \varepsilon\text{-large fibre } z.$$

Ergodicity of T is expected to force this condition (see Section 9), but the general derivability question remains open; see the open problems below. This condition is independent of (US) and is not a standing assumption here; it is used only in Corollary 6.12 below.

Lemma 6.10 (Positive-fraction balance). Let $S^* \in \mathcal{B}$ be an ε -near-maximizer of $\delta(L)$, i.e. $\mathbb{E}[\text{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \geq \delta(L) - \varepsilon$. Define the large-fibre contribution

$$\kappa := \int_{\{p_z \geq \varepsilon\}} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L(z).$$

For any $\eta \in (0, \frac{1}{2})$ with $\kappa > \eta$, the set

$$G(\eta) := \{z : p_z \geq \varepsilon, \mu_z(S^*) \mu_z((S^*)^c) \geq \eta\}$$

of ε -large balanced fibres satisfies

$$\nu_L(G(\eta)) \geq 4(\kappa - \eta).$$

No dynamical hypothesis is required.

Proof. Partition the ε -large fibres into monochromatic fibres $\mathcal{M} = \{z : p_z \geq \varepsilon, \mu_z(S^*) \mu_z((S^*)^c) < \eta\}$ and balanced fibres $G(\eta)$. On $\mathcal{M}$: $\mu_z(S^*) \mu_z((S^*)^c) < \eta$, so

$$\int_{\mathcal{M}} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L \leq \eta \int_{\mathcal{M}} p_z d\nu_L \leq \eta.$$

Hence $\int_{G(\eta)} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L \geq \kappa - \eta$. On $G(\eta)$: $\mu_z(S^*) \mu_z((S^*)^c) \leq \frac{1}{4}$ pointwise, so

$$\kappa - \eta \leq \int_{G(\eta)} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L \leq \frac{1}{4} \int_{G(\eta)} p_z d\nu_L \leq \frac{1}{4} \nu_L(G(\eta)). \quad \square$$

Lemma 6.11 (Fibre mixing bound). *Let S^* be a $(\delta(L) - \varepsilon)$ -near-maximizer of $\mathbb{E}[\text{Var}(\mathbb{1}_\cdot \mid \mathcal{O}_h^{(L)})]$ that is (c, ε) -mixing across fibres (Definition 6.9). Then*

$$(\mu \otimes \mu)(R_L) \leq \frac{1}{c} \delta(L) + C(c) \varepsilon, \quad C(c) := \frac{1+c}{c}.$$

Proof. Write $(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z)$ and split into ε -large and small-fibre parts.

Large fibres ($p_z \geq \varepsilon$). The (c, ε) -mixing condition gives $\mu_z(S^*)\mu_z((S^*)^c) \geq c p_z$ on ε -large fibres, so:

$$c \int_{\{p_z \geq \varepsilon\}} p_z^2 d\nu_L(z) \leq \int p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L(z) = \mathbb{E}[\text{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \leq \delta(L).$$

Hence $\int_{\{p_z \geq \varepsilon\}} p_z^2 d\nu_L(z) \leq \frac{1}{c} \delta(L)$.

Small fibres ($p_z < \varepsilon$). Each such fibre satisfies $p_z^2 \leq \varepsilon p_z$, so

$$\int_{\{p_z < \varepsilon\}} p_z^2 d\nu_L(z) \leq \varepsilon \int p_z d\nu_L(z) = \varepsilon.$$

Combining. Since $C(c) = \frac{1+c}{c} \geq 1$ we have $\varepsilon \leq C(c) \varepsilon$, and therefore

$$(\mu \otimes \mu)(R_L) \leq \frac{1}{c} \delta(L) + C(c) \varepsilon. \quad \square$$

The collision probability (2) defines a natural information-theoretic quantity:

$$H_2(\nu_L) := -\log((\mu \otimes \mu)(R_L)) = -\log \int p_z^2 d\nu_L(z). \quad (4)$$

This is the *Rényi-2 (collision) entropy* [31] of the delay-vector distribution.

Corollary 6.12 (Entropy characterisation of reconstruction). *Under the fibre mixing condition of Definition 6.9:*

$$\delta(L) \rightarrow 0 \iff H_2(\nu_L) \rightarrow +\infty.$$

Equivalently, reconstruction occurs if and only if the probability that two independently drawn delay vectors coincide vanishes as $L \rightarrow \infty$. Thus reconstruction is equivalent to the vanishing of collision probability $(\mu \otimes \mu)(R_L)$ introduced in Section 3.

Proof. Lemma 4.1 gives $\delta(L) \leq \frac{1}{2}(\mu \otimes \mu)(R_L)$. For the reverse, Lemma 6.11 with a $\delta(L)$ -near-maximizer S^* satisfying the (c, ε) -mixing condition gives $(\mu \otimes \mu)(R_L) \leq \frac{1}{c} \delta(L) + C(c) \varepsilon$, so $\delta(L) \rightarrow 0 \iff (\mu \otimes \mu)(R_L) \rightarrow 0$. Since $(\mu \otimes \mu)(R_L) = e^{-H_2(\nu_L)}$, this is equivalent to $H_2(\nu_L) \rightarrow +\infty$. $\square$

Thus Lemma 6.10 guarantees a positive ν_L -fraction of ε -large fibres are η -balanced without any dynamical hypothesis; the fibre mixing condition is precisely the upgrade from positive-fraction to ν_L -a.e. balance.

Remark 6.13 (Empirical entropy witness). The empirical analogue of $H_2(\nu_L)$ is

$$\hat{H}_2(\nu_L^{(n)}) := -\log \left(\frac{1}{n^2} \sum_{i,j=1}^n \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)] \right),$$

the negative log of the empirical collision probability. This is the third computable witness for reconstruction. It is *computationally simpler* than $\hat{\delta}(L, n)$: it requires no optimisation over sets, only histogram counts over delay vectors. Under fibre mixing, Corollary 6.12 shows that $\hat{L}^*$ could equivalently be defined as the first L at which $\hat{H}_2(\nu_L^{(n)})$ exceeds a threshold calibrated to n .

7 The Dynamics Theorem

7.1 Lipschitz continuity of the delay map

Lemma 7.1 (Upper Lipschitz bound). *Assume $T \in \text{Diff}^r(X)$, $h \in C^r(X)$, $r \geq 1$. Then $\Phi_h^{(L)}$ is Lipschitz:*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \leq C_L \cdot d_X(x, x') \quad \forall x, x' \in X,$$

where $C_L := \|Dh\|_{L^\infty} \cdot (\sum_{j=0}^L \|DT^j\|_{L^\infty}^2)^{1/2}$.

Proof. Component j of $\Phi_h^{(L)}$ is $h \circ T^j$. By the chain rule and compactness of X : $|h(T^j x) - h(T^j x')| \leq \|Dh\|_{L^\infty} \cdot \|DT^j\|_{L^\infty} \cdot d_X(x, x')$. Taking Euclidean norm over $j = 0, \dots, L$ gives the result. $\square$

Lemma 7.2 (Local lower Lipschitz bound). *Under (G1)–(G2) with $L \geq d-1$ (Takens regime), $T \in C^r$, $r \geq 2$: for μ -a.e. $x \in X$, there exists a neighbourhood U_x and a constant $c_x = \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ such that*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \geq c_x \cdot d_X(x, x') \quad \forall x' \in U_x.$$

Proof. By Lemma 6.2, $D\Phi_h^{(L)}|_x$ has full rank d for μ -a.e. x when $L \geq d-1$, so $\sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ μ -a.e. The lower bound on U_x follows from the inverse function theorem applied to the immersion $\Phi_h^{(L)}$ at x . $\square$

Remark 7.3 (Gap between local and global lower bound). Lemma 7.2 gives a *local* lower bound $c_x > 0$ varying over X . (G1)–(G2) guarantee $\sigma_{\min} > 0$ μ -a.e., but not on all of X . Condition (US) imposes $c := \inf_X \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, which is strictly stronger. The gap is genuine: there exist pairs (T, h) satisfying (G1)–(G2) but violating (US) on a μ -null set.

7.2 Uniform separation

For deterministic T , the conditional regularity kernel specialises to the Dirac family $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$. The TV separation pseudometric is therefore

$$d_L(x, x') := \|\Pi_h^{(L)}(x, \cdot) - \Pi_h^{(L)}(x', \cdot)\|_{\text{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')] \in \{0, 1\}.$$

Separation is binary: $d_L(x, x') = 1$ if and only if x and x' are distinguished by $\mathcal{O}_h^{(L)}$.

Theorem 7.4 (Uniform separation under (US)). *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, $L \geq d-1$. Then $\Phi_h^{(L)}$ is bi-Lipschitz on X :*

$$c \cdot d_X(x, x') \leq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \leq C_L \cdot d_X(x, x') \quad \forall x, x' \in X.$$

For any estimator $\hat{\Gamma}_h^{(n)}$ with uniform error $|\hat{\Gamma}_h^{(n)}(x) - \Phi_h^{(L)}(x)| \leq \varepsilon_n$ uniformly in x , the empirical separation $\hat{d}_L(x, x') := |\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')|$ satisfies

$$\hat{d}_L(x, x') \geq c \cdot d_X(x, x') - 2\varepsilon_n.$$

In particular, if $d_L(x, x') = 1$ (i.e. $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$), then $\hat{d}_L(x, x') > 0$ for all n large enough that $\varepsilon_n < c \cdot d_X(x, x')/2$.

Proof. Upper bound: Lemma 7.1. Lower bound: (US) directly gives $|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \geq c \cdot d_X(x, x')$ for all $x, x' \in X$. For the empirical bound: by the triangle inequality, $|\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')| \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| - 2\varepsilon_n \geq c \cdot d_X(x, x') - 2\varepsilon_n$. If $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ then by bi-Lipschitz $d_X(x, x') \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')|/C_L > 0$, so the right side is eventually positive. $\square$

Theorem 7.5 (Dynamics Theorem). *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, μ smooth positive density on compact d -manifold X , $L \geq d - 1$. Let $\hat{\Gamma}_h^{(n)}$ be the empirical delay map estimator with uniform error $\varepsilon_n = C''n^{-\beta/(2\beta+d)}$ (Proposition 5.6). Then:*

- (a) (Convergence) $\|\hat{\Gamma}_h^{(n)} - \Phi_h^{(L)}\|_{L^\infty} \rightarrow 0$ at rate $n^{-\beta/(2\beta+d)}$.
- (b) (Separation) For $\mu \otimes \mu$ -a.e. (x, x') with $x \neq x'$: $\hat{d}_L(x, x') > 0$ for all large n .
- (c) (Scope) Condition (US) is not a consequence of (G1)–(G2). Without (US), the kernel witness $\hat{d}_L$ can vanish on a positive-measure set of pairs even when $d_L = 1$ and reconstruction holds.

Proof. (a) Proposition 5.6 with $d_{\text{eff}}(L) = d$ gives $\varepsilon_n \rightarrow 0$ at rate $n^{-\beta/(2\beta+d)}$.

(b) Since μ is non-atomic (smooth positive density), $\mu \otimes \mu$ almost every pair satisfies $d_X(x, x') > 0$. For such pairs $d_L(x, x') = 1$ (reconstruction and injectivity of $\Phi_h^{(L)}$), and by Theorem 7.4, $\hat{d}_L(x, x') \geq c \cdot d_X(x, x') - 2\varepsilon_n$. For fixed $x \neq x'$, $c \cdot d_X(x, x') > 0$ and $\varepsilon_n \rightarrow 0$, so the right side is eventually positive.

(c) The proof of Theorem 7.4 uses (US) in the lower bound. Without it, there exist points x where $\sigma_{\min}(D\Phi_h^{(L)}|_x) \rightarrow 0$ along sequences $x'_n \rightarrow x$, and $|\Phi_h^{(L)}(x'_n) - \Phi_h^{(L)}(x)|/d_X(x, x'_n) \rightarrow 0$. Any estimator inherits this degeneracy. $\square$

8 The Conjunction Theorem

This section formalises the complete reconstruction diagnostic and proves that both witnesses certify the same event when reconstruction and (US) hold.

8.1 The honest bridge

The delay vector $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$ is a trajectory of the Markov chain on h -space with one-step kernel Π_h . The σ -algebra $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ is exactly the information generated by L steps of this Markov chain starting from x . This gives an algebraic identity:

$$\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)},$$

which is not an approximation but a definition. Consequently, the σ -algebra separation event and the metric separation event are identical:

Lemma 8.1 (Algebra separation equals metric separation). *For deterministic T and $\mu \otimes \mu$ -a.e. (x, x') :*

$$d_L(x, x') = 1 \iff x \text{ and } x' \text{ are separated by } \mathcal{O}_h^{(L)}.$$

These are the same event, not merely correlated.

Proof. For deterministic T : $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$, so $d_L(x, x') = \|\delta_{\Phi_h^{(L)}(x)} - \delta_{\Phi_h^{(L)}(x')}\|_{\text{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')]$. And x is separated from x' by $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ if and only if $\exists E \in \mathcal{O}_h^{(L)}$ with $\mathbb{1}_E(x) \neq \mathbb{1}_E(x')$, i.e., if and only if $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ (since $\mathcal{O}_h^{(L)}$ is generated by the level sets of $\Phi_h^{(L)}$). The two conditions are identical. $\square$

Under reconstruction ($\delta(L) = 0$, equivalently $\Phi_h^{(L)}$ injective μ -a.e.) and (US), injectivity holds everywhere, so $d_L(x, x') = 1$ for $\mu \otimes \mu$ -a.e. $x \neq x'$.

8.2 The joint diagnostic

Definition 8.2 (Joint reconstruction diagnostic). Fix tolerances $\tau_n, \eta_n > 0$. Define

$$\text{Reconstruct}(L, n) := \{\hat{\delta}(L, n) \leq \tau_n\} \wedge \{\hat{d}_L(x, x') > \eta_n \text{ for } \mu \otimes \mu\text{-most pairs}\}.$$

Theorem 8.3 (Conjunction Theorem). *Assume the hypotheses of Theorem 6.8 (for the algebra side) and Theorem 7.5 (for the dynamics side). Set $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$ and $\eta_n = \sqrt{\varepsilon_n}$ where $\varepsilon_n = C' n^{-2s/(2s+d)}$. Then:*

(a) (Conjunction fires under reconstruction + (US)). *Under $\delta(L) = 0$ for $L \geq L^*(n)$ and (US):*

$$P(\text{Reconstruct}(\hat{L}^*, n)) \rightarrow 1 \quad \text{exponentially in } n.$$

(b) (Algebra witness fails when reconstruction fails). *If $\delta(L) > 0$ for all L : $P(\hat{\delta}(L, n) \leq \tau_n) \rightarrow 0$ for every fixed L , and $\hat{L}^* \rightarrow \infty$.*

(c) (Kernel witness fails when (US) fails). *If $\sigma_{\min}(D\Phi_h^{(L)}|_x) \rightarrow 0$ on a positive-measure set S : $P(\hat{d}_L > \eta_n \text{ for } \mu \otimes \mu\text{-most pairs}) \rightarrow 0$ even when $\hat{\delta}(L, n) \rightarrow 0$.*

In particular: the conjunction is not achievable when either hypothesis fails.

Proof. Part (a). By Theorem 6.6, $P(\hat{L}^* = L^*(n)) \rightarrow 1$ exponentially. On $\{\hat{L}^* = L^*(n)\}$: (i) $\hat{\delta}(\hat{L}^*, n) \leq \tau_n$ whp by Theorem 5.7 and the definition of $L^*(n)$ ($\delta(L^*(n)) = 0$ under reconstruction). (ii) By Theorem 7.5(b), $\hat{d}_{\hat{L}^*}(x, x') > 0$ for $\mu \otimes \mu$ -a.e. (x, x') with $x \neq x'$, eventually. The set $\{d_X(x, x') \leq (2\varepsilon_n + \eta_n)/c\}$ has $\mu \otimes \mu$ -measure $\rightarrow 0$ as $\eta_n/c \rightarrow 0$, so $\hat{d}_{\hat{L}^*} > \eta_n$ for $\mu \otimes \mu$ -most pairs.

Part (b). If $\delta(L) > 0$ for all L , then by Theorem 5.7, $\hat{\delta}(L, n) \geq \delta(L) - \varepsilon_n \geq \delta(L)/2 > \tau_n$ with high probability for all fixed L , so the first condition of Reconstruct never fires.

Part (c). If $S = \{x : \sigma_{\min}(D\Phi_h^{(L)}|_x) = 0\}$ has $\mu(S) > 0$, then for $x \in S$ there exist sequences $x'_k \rightarrow x$ with $|\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)|/d_X(x'_k, x) \rightarrow 0$. Any consistent estimator satisfies $\hat{d}_L(x, x'_k) \leq |\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)| + 2\varepsilon_n \rightarrow 0$. The set of pairs where $\hat{d}_L \leq \eta_n$ thus has positive $\mu \otimes \mu$ -measure, and the kernel condition fails. $\square$

9 Discussion

The results of this paper divide into two layers. Section 2 establishes general observational certification: the population separation defect $\delta(\mathcal{G})$ converges to zero under CE and honest refinement, the empirical U-statistic proxy concentrates under IID sampling, and a persistent empirical floor is evidence against CE. These results require no temporal or geometric structure. Sections 6–8 establish the dynamical sharpening: honest refinement is automatic under lag growth, and mixing upgrades consistency to minimax-optimal rates with computable witnesses. The open problems below all concern the dynamical layer; the general layer is complete at the level of consistency.

Polynomial mixing. The Algebra Theorem (Theorem 6.8) and the Elbow Location Theorem (Theorem 6.6) require exponential mixing with resolvability constant $C_\lambda > 4$. Under polynomial mixing $\delta(L) \lesssim L^{-\alpha}$, the elbow drop at $L^*(n)$ satisfies $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \rightarrow 0$ relative to ε_n : the elbow and the noise floor merge asymptotically and cannot be separated. A different stopping criterion (e.g. integrated elbow, or a bandwidth-adaptive rule) is needed.

The $d > 2s$ hypothesis. Proposition 5.2 requires $d > 2s$ for the Rademacher concentration to be useful at rate ε_n . When $d \leq 2s$, localised Rademacher complexity (Bartlett–Bousquet–Mendelson [2]) gives adaptive concentration without this condition but requires additional work.

Uniform separation from first principles. Condition (US) is a hypothesis in Theorems 7.5 and 8.3. The natural conjecture is that T Anosov with h generic implies (US) via the uniform hyperbolicity constants. This would close the gap between (G1)–(G2) and (US) in the hyperbolic category.

Remark 9.1 (Observable separation and Lyapunov exponents). The bi-Lipschitz property of $\Phi_h^{(L)}$ established in Theorem 7.5 under (G1)–(G2)+(US) has a direct bearing on the observable separation exponent introduced in [27] (Remark II.7). Recall that the *observable separation exponent* is

$$\lambda_h(x, y) := \limsup_{n \rightarrow \infty} \frac{1}{n} \log d(\Phi_h^{(L)}(T^n x), \Phi_h^{(L)}(T^n y)),$$

measuring how fast the observable distinguishes initial conditions under iteration. The bi-Lipschitz sandwich $c d(x, y) \leq d(\Phi_h^{(L)}(x), \Phi_h^{(L)}(y)) \leq C d(x, y)$, valid μ -a.e. under (G1)–(G2)+(US), implies

$$\lambda_h(x, y) = \lambda_1(x, y) \quad \mu\text{-a.e.},$$

where λ_1 is the top Lyapunov exponent of T in the ambient metric. That is, under faithful reconstruction and uniform separation, the observable separation rate and the geometric instability rate coincide. This identifies the observable separation exponent as a computable proxy for λ_1 accessible directly from time-series data, without knowledge of the tangent bundle or the derivative cocycle.

Stochastic T . For stochastic dynamics, $\Pi_h^{(L)}(x, \cdot)$ is not a Dirac delta. The TV separation d_L is no longer binary, and Hölder continuity of $x \mapsto \Pi_h^{(L)}(x, \cdot)$ in TV is both necessary and non-trivial. The four-link chain of Proposition 5.6 adapts, but the Conjunction Theorem’s proof structure changes substantially.

Sharp rates for the data-driven lag. Theorem 6.6 shows $P(\hat{L}^* = L^*(n)) \rightarrow 1$ but does not bound the fluctuations of $\hat{L}^*$ around $L^*(n)$ on the event $\{\hat{L}^* \neq L^*(n)\}$. A Berry–Esseen-type result for $\hat{L}^* - L^*(n)$ would sharpen the stopping rule.

Concentration of the entropy witness. The empirical collision entropy $\hat{H}_2(\nu_L^{(n)})$ is a computationally simpler stopping criterion (Remark 6.13). Asymptotic equivalence with $\hat{\delta}(L, n)$ is established under fibre mixing (Corollary 6.12), but a finite- n concentration result for $\hat{H}_2(\nu_L^{(n)})$ — comparable to Proposition 5.2 for $\hat{\delta}(L, n)$ — is not proved here. Such a result requires only McDiarmid-type bounds on the U-statistic $\frac{1}{n^2} \sum_{i,j} \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)]$, with no new structural assumptions beyond those already made.

Derivability of fibre mixing (the main conceptual open problem). The central open question is whether the fibre mixing condition (Definition 6.9) is derivable from dynamical hypotheses, or whether it is an irreducible admissibility condition analogous to CE in [28].

Lemma 6.10 settles Step A: a positive ν_L -fraction of ε -large fibres are η -balanced, with an explicit lower bound depending only on the large-fibre contribution κ and not on any dynamical hypothesis.

The open question is Step B: what is the weakest dynamical condition that upgrades positive-fraction balance to ν_L -a.e. balance? Ergodicity is the natural candidate: in an ergodic system, a near-maximizer of $\delta(L)$ must cross fibres rather than align with them, since aligned events are $\mathcal{O}_h^{(L)}$ -measurable and thus carry zero approximation error. The informal argument is clear, but formalizing the connection between ergodicity and the a.e. balance condition is the isolated open gap. If ergodicity suffices, the fibre mixing condition is derivable and the structural analogy with CE breaks; if it does not, the analogy is complete.

References

- [1] Roy L. Adler. Symbolic dynamics and Markov partitions. *Bull. Amer. Math. Soc. (N.S.)*, 35(1):1–57, 1998.
- [2] Peter L. Bartlett, Olivier Bousquet, and Shahar Mendelson. Local Rademacher complexities. *Annals of Statistics*, 33(4):1497–1537, 2005.
- [3] Peter L. Bartlett and Shahar Mendelson. Rademacher and Gaussian complexities: risk bounds and structural results. *Journal of Machine Learning Research*, 3:463–482, 2002.
- [4] David Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953.
- [5] Vladimir I. Bogachev. *Measure Theory, Volumes I–II*. Springer, 2007.
- [6] Rodrigo Cardona, Diego Alejandro Mejía, and Iván Uribe-Zapata. Finitely additive measures on Boolean algebras. arXiv preprint arXiv:2503.08910, 2025.
- [7] J. R. Choksi. Inverse limits of measure spaces. *Proceedings of the London Mathematical Society*, 8(3):321–342, 1958.
- [8] Joseph L. Doob. *Stochastic Processes*. Wiley, New York, 1953.
- [9] Eugene B. Dynkin. *Markov Processes*. Springer, 1965.
- [10] Ronald Fagin, Joseph Y. Halpern, and Nimrod Megiddo. A logic for reasoning about probabilities. *Information and Computation*, 87(1–2):78–128, 1990.
- [11] D. H. Fremlin. *Measure Theory*, volume 1. Torres Fremlin, 2003.
- [12] László Györfi, Michael Kohler, Adam Krzyżak, and Harro Walk. *A Distribution-Free Theory of Nonparametric Regression*. Springer, 2002.
- [13] Paul R. Halmos. In general a measure preserving transformation is mixing. *Annals of Mathematics*, 45(4):786–792, 1944.
- [14] Paul R. Halmos. *Measure Theory*. Van Nostrand, 1950.
- [15] Paul R. Halmos and Leonard J. Savage. Application of the Radon–Nikodym theorem to the theory of sufficient statistics. *Annals of Mathematical Statistics*, 20(2):225–241, 1949.
- [16] Wassily Hoeffding. A class of statistics with asymptotically normal distribution. *Annals of Mathematical Statistics*, 19(3):293–325, 1948.
- [17] Douglas N. Hoover. Probability logic. *Annals of Mathematical Logic*, 14(3):287–313, 1978.
- [18] Herbert Jaeger. Observable operator models for discrete stochastic time series. Technical Report GMD Report 42, German National Research Center for Information Technology, 2000.
- [19] Peter T. Johnstone. *Stone Spaces*, volume 3 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1982.
- [20] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.
- [21] Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.

- [22] Michel Ledoux and Michel Talagrand. *Probability in Banach Spaces*. Springer, 1991.
- [23] Michael L. Littman, Richard S. Sutton, and Satinder P. Singh. Predictive representations of state. In *Advances in Neural Information Processing Systems 14*, pages 1555–1561. MIT Press, 2001.
- [24] Jerzy Łoś. Quelques remarques, théorèmes et problèmes sur les classes définissables d’algèbres. In *Mathematical Interpretation of Formal Systems*, pages 98–113. North-Holland, 1955.
- [25] Patrick S. Mahon. Certifying reconstruction from finite data. Preprint, 2026.
- [26] Patrick S. Mahon. Countable additivity is not first-order axiomatizable. Preprint, 2026.
- [27] Patrick S. Mahon. Dynamics and reconstruction in the observable measure: Conditional regularity, Koopman–Perron duality, and the density bridge. Preprint, 2026.
- [28] Patrick S. Mahon. Probability from observation: Carathéodory and Stone routes to the observable extension theorem. Preprint, 2026.
- [29] Colin McDiarmid. On the method of bounded differences. In *Surveys in Combinatorics 1989*, volume 141 of *London Mathematical Society Lecture Note Series*, pages 148–188. Cambridge University Press, 1989.
- [30] K. P. S. Bhaskara Rao and M. Bhaskara Rao. *Theory of Charges: A Study of Finitely Additive Measures*. Academic Press, New York, 1983.
- [31] Alfréd Rényi. On measures of entropy and information. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 547–561. University of California Press, 1961.
- [32] Vladimir Abramovich Rokhlin. On the fundamental ideas of measure theory. *American Mathematical Society Translations*, 10:1–54, 1952.
- [33] Tim Sauer, James A. Yorke, and Martin Casdagli. Embedology. *Journal of Statistical Physics*, 65(3–4):579–616, 1991.
- [34] Stephen Smale. An infinite dimensional version of Sard’s theorem. *American Journal of Mathematics*, 87(4):861–866, 1965.
- [35] Charles J. Stone. Optimal global rates of convergence for nonparametric regression. *Annals of Statistics*, 10(4):1040–1053, 1982.
- [36] Marshall Harvey Stone. The theory of representations for Boolean algebras. *Transactions of the American Mathematical Society*, 40(1):37–111, 1936.
- [37] Floris Takens. Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick 1980*, volume 898 of *Lecture Notes in Mathematics*, pages 366–381. Springer, 1981.
- [38] Kôzaku Yosida and Edwin Hewitt. Finitely additive measures. *Transactions of the American Mathematical Society*, 72(1):46–66, 1952.